Informatica® Axon Data Governance
7.3.8

Installation Guide

Informatica Axon Data Governance Installation Guide
7.3.8
September 2026

© Copyright Informatica LLC 2013, 2026

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

Portions of this software and/or documentation are subject to copyright held by third parties, including without limitation: Copyright DataDirect Technologies. All rights reserved. Copyright © Sun Microsystems. All rights reserved. Copyright © RSA Security Inc. All Rights Reserved. Copyright © Ordinal Technology Corp. All rights reserved. Copyright © Aandacht c.v. All rights reserved. Copyright Genivia, Inc. All rights reserved. Copyright Isomorphic Software. All rights reserved. Copyright © Meta Integration Technology, Inc. All rights reserved. Copyright © Intalio. All rights reserved. Copyright © Oracle. All rights reserved. Copyright © Adobe Systems Incorporated. All rights reserved. Copyright © DataArt, Inc. All rights reserved. Copyright © ComponentSource. All rights reserved. Copyright © Microsoft Corporation. All rights reserved. Copyright © Rogue Wave Software, Inc. All rights reserved. Copyright © Teradata Corporation. All rights reserved. Copyright © Yahoo! Inc. All rights reserved. Copyright © Glyph & Cog, LLC. All rights reserved. Copyright © Thinkmap, Inc. All rights reserved. Copyright © Clearpace Software Limited. All rights reserved. Copyright © Information Builders, Inc. All rights reserved. Copyright © OSS Nokalva, Inc. All rights reserved. Copyright Edifecs, Inc. All rights reserved. Copyright Cleo Communications, Inc. All rights reserved. Copyright © International Organization for Standardization 1986. All rights reserved. Copyright © ej-technologies GmbH. All rights reserved. Copyright © Jaspersoft Corporation. All rights reserved. Copyright © International Business Machines Corporation. All rights reserved. Copyright © yWorks GmbH. All rights reserved. Copyright © Lucent Technologies. All rights reserved. Copyright © University of Toronto. All rights reserved. Copyright © Daniel Veillard. All rights reserved. Copyright © Unicode, Inc. Copyright IBM Corp. All rights reserved. Copyright © MicroQuill Software Publishing, Inc. All rights reserved. Copyright © PassMark Software Pty Ltd. All rights reserved. Copyright © LogiXML, Inc. All rights reserved. Copyright © 2003-2010 Lorenzi Davide, All rights reserved. Copyright © Red Hat, Inc. All rights reserved. Copyright © The Board of Trustees of the Leland Stanford Junior University. All rights reserved. Copyright © EMC Corporation. All rights reserved. Copyright © Flexera Software. All rights reserved. Copyright © Jinfonet Software. All rights reserved. Copyright © Apple Inc. All rights reserved. Copyright © Telerik Inc. All rights reserved. Copyright © BEA Systems. All rights reserved. Copyright © PDFlib GmbH. All rights reserved. Copyright © Orientation in Objects GmbH. All rights reserved. Copyright © Tanuki Software, Ltd. All rights reserved. Copyright © Ricebridge. All rights reserved. Copyright © Sencha, Inc. All rights reserved. Copyright © Scalable Systems, Inc. All rights reserved. Copyright © jQWidgets. All rights reserved. Copyright © Tableau Software, Inc. All rights reserved. Copyright© MaxMind, Inc. All Rights Reserved. Copyright © TMate Software s.r.o. All rights reserved. Copyright © MapR Technologies Inc. All rights reserved. Copyright © Amazon Corporate LLC. All rights reserved. Copyright © Highsoft. All rights reserved. Copyright © Python Software Foundation. All rights reserved. Copyright © BeOpen.com. All rights reserved. Copyright © CNRI. All rights reserved.

This product includes software developed by the Apache Software Foundation (http://www.apache.org/), and/or other software which is licensed under various versions of the Apache License (the "License"). You may obtain a copy of these Licenses at http://www.apache.org/licenses/. Unless required by applicable law or agreed to in writing, software distributed under these Licenses is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the Licenses for the specific language governing permissions and limitations under the Licenses.

This product includes software which was developed by Mozilla (http://www.mozilla.org/), software copyright The JBoss Group, LLC, all rights reserved; software copyright © 1999-2006 by Bruno Lowagie and Paulo Soares and other software which is licensed under various versions of the GNU Lesser General Public License Agreement, which may be found at http:// www.gnu.org/licenses/lgpl.html. The materials are provided free of charge by Informatica, "as-is", without warranty of any kind, either express or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose.

The product includes ACE(TM) and TAO(TM) software copyrighted by Douglas C. Schmidt and his research group at Washington University, University of California, Irvine, and Vanderbilt University, Copyright (©) 1993-2006, all rights reserved.

This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (copyright The OpenSSL Project. All Rights Reserved) and redistribution of this software is subject to terms available at http://www.openssl.org and http://www.openssl.org/source/license.html.

This product includes Curl software which is Copyright 1996-2013, Daniel Stenberg, <daniel@haxx.se>. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://curl.haxx.se/docs/copyright.html. Permission to use, copy, modify, and distribute this software for any purpose with or without fee is hereby granted, provided that the above copyright notice and this permission notice appear in all copies.

The product includes software copyright 2001-2005 (©) MetaStuff, Ltd. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://www.dom4j.org/ license.html.

The product includes software copyright © 2004-2007, The Dojo Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://dojotoolkit.org/license.

This product includes ICU software which is copyright International Business Machines Corporation and others. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://source.icu-project.org/repos/icu/icu/trunk/license.html.

This product includes software copyright © 1996-2006 Per Bothner. All rights reserved. Your right to use such materials is set forth in the license which may be found at http:// www.gnu.org/software/ kawa/Software-License.html.

This product includes OSSP UUID software which is Copyright © 2002 Ralf S. Engelschall, Copyright © 2002 The OSSP Project Copyright © 2002 Cable & Wireless Deutschland. Permissions and limitations regarding this software are subject to terms available at http://www.opensource.org/licenses/mit-license.php.

This product includes software developed by Boost (http://www.boost.org/) or under the Boost software license. Permissions and limitations regarding this software are subject to terms available at http:/ /www.boost.org/LICENSE_1_0.txt.

This product includes software copyright © 1997-2007 University of Cambridge. Permissions and limitations regarding this software are subject to terms available at http:// www.pcre.org/license.txt.

This product includes software copyright © 2007 The Eclipse Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http:// www.eclipse.org/org/documents/epl-v10.php and at http://www.eclipse.org/org/documents/edl-v10.php.

This product includes software licensed under the terms at http://www.tcl.tk/software/tcltk/license.html, http://www.bosrup.com/web/overlib/?License, http:// www.stlport.org/doc/ license.html, http://asm.ow2.org/license.html, http://www.cryptix.org/LICENSE.TXT, http://hsqldb.org/web/hsqlLicense.html, http:// httpunit.sourceforge.net/doc/ license.html, http://jung.sourceforge.net/license.txt , http://www.gzip.org/zlib/zlib_license.html, http://www.openldap.org/software/ release/license.html, http://www.libssh2.org, http://slf4j.org/license.html, http://www.sente.ch/software/OpenSourceLicense.html, http://fusesource.com/downloads/ license-agreements/fuse-message-broker-v-5-3- license-agreement; http://antlr.org/license.html; http://aopalliance.sourceforge.net/; http://www.bouncycastle.org/ licence.html; http://www.jgraph.com/jgraphdownload.html; http://www.jcraft.com/jsch/LICENSE.txt; http://jotm.objectweb.org/bsd_license.html; . http://www.w3.org/ Consortium/Legal/2002/copyright-software-20021231; http://www.slf4j.org/license.html; http://nanoxml.sourceforge.net/orig/copyright.html; http://www.json.org/ license.html; http://forge.ow2.org/projects/javaservice/, http://www.postgresql.org/about/licence.html, http://www.sqlite.org/copyright.html, http://www.tcl.tk/ software/tcltk/license.html, http://www.jaxen.org/faq.html, http://www.jdom.org/docs/faq.html, http://www.slf4j.org/license.html; http://www.iodbc.org/dataspace/ iodbc/wiki/iODBC/License; http://www.keplerproject.org/md5/license.html; http://www.toedter.com/en/jcalendar/license.html; http://www.edankert.com/bounce/

index.html; http://www.net-snmp.org/about/license.html; http://www.openmdx.org/#FAQ; http://www.php.net/license/3_01.txt; http://srp.stanford.edu/license.txt; http://www.schneier.com/blowfish.html; http://www.jmock.org/license.html; http://xsom.java.net; http://benalman.com/about/license/; https://github.com/CreateJS/EaselJS/blob/master/src/easeljs/display/Bitmap.js; http://www.h2database.com/html/license.html#summary; http://jsoncpp.sourceforge.net/LICENSE; http://jdbc.postgresql.org/license.html; http://protobuf.googlecode.com/svn/trunk/src/google/protobuf/descriptor.proto; https://github.com/rantav/hector/blob/master/LICENSE; http://web.mit.edu/Kerberos/krb5-current/doc/mitK5license.html; http://jibx.sourceforge.net/jibx-license.html; https://github.com/lyokato/libgeohash/blob/master/LICENSE; https://github.com/hjiang/jsonxx/blob/master/LICENSE; https://code.google.com/p/lz4/; https://github.com/jedisct1/libsodium/blob/master/LICENSE; http://one-jar.sourceforge.net/index.php?page=documents&file=license; https://github.com/EsotericSoftware/kryo/blob/master/license.txt; http://www.scala-lang.org/license.html; https://github.com/tinkerpop/blueprints/blob/master/LICENSE.txt; http://gee.cs.oswego.edu/dl/classes/EDU/oswego/cs/dl/util/concurrent/intro.html; https://aws.amazon.com/asl/; https://github.com/twbs/bootstrap/blob/master/LICENSE; https://sourceforge.net/p/xmlunit/code/HEAD/tree/trunk/LICENSE.txt; https://github.com/documentcloud/underscore-contrib/blob/master/LICENSE, and https://github.com/apache/hbase/blob/master/LICENSE.txt.

This product includes software licensed under the Academic Free License (http://www.opensource.org/licenses/afl-3.0.php), the Common Development and Distribution License (http://www.opensource.org/licenses/cddl1.php) the Common Public License (http://www.opensource.org/licenses/cpl1.0.php), the Sun Binary Code License Agreement Supplemental License Terms, the BSD License (http:// www.opensource.org/licenses/bsd-license.php), the new BSD License (http://opensource.org/licenses/BSD-3-Clause), the MIT License (http://www.opensource.org/licenses/mit-license.php), the Artistic License (http://www.opensource.org/licenses/artistic-license-1.0) and the Initial Developer's Public License Version 1.0 (http://www.firebirdsql.org/en/initial-developer-s-public-license-version-1-0/).

This product includes software copyright © 2003-2006 Joe Walnes, 2006-2007 XStream Committers. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://xstream.codehaus.org/license.html. This product includes software developed by the Indiana University Extreme! Lab. For further information please visit http://www.extreme.indiana.edu/.

This product includes software Copyright (c) 2013 Frank Balluffi and Markus Moeller. All rights reserved. Permissions and limitations regarding this software are subject to terms of the MIT license.

See patents at https://www.informatica.com/legal/patents.html.

DISCLAIMER: Informatica LLC provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica LLC does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-09-22

Table of Contents

Preface 7

Informatica Resources 7

- Informatica Network 7
- Informatica Knowledge Base 7
- Informatica Documentation 7
- Informatica Product Availability Matrices 8
- Informatica Marketplace 8
- Informatica Global Customer Support 8

Chapter 1: Installation Overview 9

Installation Overview 9

Installation Process 9

Users 10

- Users in the Linux Environment 11
- User Profiles in Axon Data Governance 11
- User Profiles in Axon Data Marketplace 13

Chapter 2: Before You Run the Installer 14

System Requirements 14

- Port Requirements 16
- RPM Package Manager Files 18
- Permissions for Running RPM Commands 23
- Update Cryptographic Policy 24

Tasks Before You Run the Installer 24

- Create an Axon Administrator User 24
- Create Directories 24
- Provide Permissions for Running RPM Commands 25
- Provide Permissions to the Objects Database 25
- Configure Port Access 25
- Verify Host Name and IP Address Configuration 25
- Configure Host Names for Related Components 25
- Configure Clusters for Graph Database 26
- Configure ulimit Values 26
- Install Language Fonts 26
- Verify the Axon License Key 26
- Download Compatible Versions of Elasticsearch and Logstash Files 26
- Verify License for Data Quality Rule Automation 27
- Extract the Installer Files 27
- Run the Validation Utility 27
- Obtain Domain Details 28

Obtain SSL Details. . . . 29
Create a New Database Schema to Automate Data Quality Rules. . . . 29

Chapter 3: Installer Directory Structure. . . . 30

Chapter 4: Running the Installer. . . . 31

Running the Installer Overview. . . . 31
Running the Installer in Console Mode. . . . 31
Installing Axon. . . . 32
Installing the Axon Agent. . . . 37
Installing Axon and the Axon Agent. . . . 39
Running the Installer in Silent Mode. . . . 47

Chapter 5: After You Run the Installer. . . . 48

Configure SSL Certificates for the Message Broker Service. . . . 48
Disable or Re-enable Segmentation. . . . 49
Configure Access to Enterprise Data Catalog. . . . 49
Connect to an SSL-Enabled Enterprise Data Catalog Instance. . . . 51
Configure Access to Informatica Data Quality. . . . 52
Configure Access to Data Privacy Management. . . . 52
Configure Axon with Operating System Profiles Enabled User for Data Integration Service. . . . 53
Connect to External Change Request Systems. . . . 54
Configure Additional Fields for External Change Request System. . . . 56
Update Cache with Configuration Values. . . . 59
Configure SSL-certified User Authority for Axon. . . . 59
Check the Status of the Services. . . . 60
Log In To Axon. . . . 61
Apache Log4j RCE Vulnerabilities for Axon. . . . 62

Chapter 6: Uninstallation. . . . 63

Running the Axon Uninstaller. . . . 63
Running the Axon Agent Uninstaller. . . . 64

Chapter 7: Troubleshooting. . . . 65

Troubleshooting Overview. . . . 65
Services Fail to Start After Installation. . . . 65
Cannot Log In to Axon. . . . 66
Cannot Start the HTTPD Service because SSL is Password-Protected. . . . 67
Axon Does Not Load in the Browser. . . . 68
Incorrect Host Name. . . . 69
Validation Utility Errors. . . . 69
Uninstallation is Incomplete. . . . 70
Warnings in Configure.log Files. . . . 70

Installation errors. 70

Index. 72

Preface

Follow the instructions in the *Axon Data Governance Installation Guide* to install Axon in your enterprise. This guide covers the tasks that the Axon system administrator can perform using the Axon installer.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

Installation Overview

This chapter includes the following topics:

- Installation Overview, 9
- Installation Process, 9
- Users, 10

Installation Overview

Informatica provides a validation utility to verify if your system meets the software requirements for installation, and an installer to install Axon. Use the installer to install Axon, the Axon Agent, or both of them.

The Axon Agent is required by Axon to connect to Informatica Data Quality. You can install the Axon Agent on the same machine where you install Axon, or on a different machine. You must provide the details of your Informatica domain when you install the Axon Agent.

The Informatica domain is a collection of nodes and services that define the basic administrative unit for Informatica Data Quality.

Installation Process

The installation of Axon consists of multiple phases that include preparation for the installation, planning the deployment type, and post-installation tasks.

The Axon installation consists of the following phases:

Preparation

Set up the system to meet the operating system requirements to ensure that you can successfully install and run Axon and the Axon Agent. Run the validation utility to verify if your system meets the requirements.

Planning

Before you run the Axon installer, you must plan your deployment.

Integrating Data Quality. Determine whether you intend to retrieve data quality scores from Informatica Data Quality and display them in Axon dashboards. If yes, you must install both Axon and the Axon Agent. You can install the Axon Agent on the same machine where you install Axon or on a different machine.

Retrieving Enterprise Metadata. Determine whether you want to discover and retrieve enterprise metadata from Enterprise Data Catalog. To integrate Axon with Enterprise Data Catalog, you need to configure certain advanced settings after you install Axon.

Complying with Data Security Standards. Determine whether you want Axon to integrate with Data Privacy Management to verify if your data meets the security standards that are required for your organization. To integrate Axon with Data Privacy Management, you need to configure certain advanced settings after you install Axon.

The following table lists the choices that the Axon installer provides. Plan your deployment based on these choices before you run the installer:

Choice	Planning	Task
Integration with Informatica Data Quality. Axon integrates with Informatica Data Quality to retrieve data quality scores and display them in the Axon dashboards.	Keep the details of your Informatica domain ready.	Choose to install Axon and the Axon Agent, or install only the Axon Agent if you have already installed Axon.
Integration with Informatica Enterprise Data Catalog. Axon integrates with Enterprise Data Catalog to discover and retrieve metadata from the catalog platform. The catalog represents an indexed inventory of all the data assets in the enterprise.	Keep the details of your Enterprise Data Catalog deployment ready.	Choose the advanced configuration step during Axon installation.
Integration with Informatica Data Privacy Management. Axon integrates with Data Privacy Management to identify sensitive data in your organization and verify if they comply with data security standards.	Keep the details of your Data Privacy Management deployment ready.	Choose the advanced configuration step during Axon installation.

To see the versions of other Informatica products that are supported by Axon, see the Product Availability Matrix on the Informatica Network.

Before you run the Axon installer, perform the preparatory steps described in the *Before You Run the Installer* chapter.

After Installation

After you run the installer, perform the tasks described in the *After You Run the Installer* chapter. Access Axon by typing the fully qualified host name in a web browser. Log in to verify if the installation is successful.

Users

Users in Axon have privileges that allow them to interact with the Axon application in different ways. To install and manage Axon effectively, you must have users with the correct privileges in the Linux environment and Axon interface.

Users in the Linux Environment

The installer file installs Axon in a home directory that is designated for the installer binaries and application files. This directory must be managed by a non-root user.

To install and manage Axon, you must have the following users in the Linux environment:

Root User

The Linux root user creates the Axon Administrator in Linux and provides a home directory to install Axon.

Axon Administrator

This user is a non-root user who installs and manages Axon. The Axon Administrator monitors the Axon application in the Linux environment, troubleshoots Axon services that might have stopped, and configures Axon in Linux to connect with other applications.

The Axon Administrator must have read and write permissions on the Axon installation directory. The Axon Administrator does not require the root or sudo privilege.

User Profiles in Axon Data Governance

Every Axon Data Governance user has a user profile. The type of profile decides the tasks that a user can perform in Axon. An administrator associates user profiles to user accounts.

Users can have one of the following profiles associated with their account:

WebUser

A WebUser is the most basic user profile in Axon. WebUsers have permission to view content.

An administrator can provide advanced permissions to a WebUser on a facet by assigning a role. SuperAdmin users can adjust the permissions applicable to a role. Any user with the Edit privileges for an object can assign roles to WebUsers. The WebUser then becomes a stakeholder for the facet and can perform other tasks to create or manage content for the facet. Stakeholder roles can provide New permissions for a facet and Edit permissions for individual objects. For example, a WebUser with the relevant role and permission can bulk upload data to Axon. A WebUser with the New permission on a Glossary facet can create Glossary objects.

If a SuperAdmin user hides non-public objects from Unison search, the non-public objects do not appear in Unison search results for a WebUser that has no stakeholder role on any object of the facet type.

Admin

Users can perform administrative tasks on facets if they are assigned the Admin profile. Users with the admin profile have access to the Admin Panel.

Users with the Admin profile can perform the following tasks from the Admin Panel:

- Automate data quality rules
- Bulk update objects from the Unison grid
- Bulk upload objects
- Configure default values for mandatory change requests
- Configure saved searches
- Create custom fields for facet objects
- Download logs
- Link Enterprise Data Catalog resources to systems and fields to attributes
- Manage locks

- Send Axon diagnostic information to Informatica Global Customer Support
- View licensed users
- View non-public and deleted objects
- View roles and permissions
- Import migrated data
- Use the static page editor

SuperAdmin

Users with the SuperAdmin profile can perform all administrative tasks in Axon. In addition to the tasks that a user with the Admin profile can perform, a user with a SuperAdmin profile can perform the following tasks:

- Assign roles and permissions
- Create segments and assign users to the segments
- Change the organization logo in the toolbar and the colors of the Axon interface
- Configure access control for users
- Configure audit history period
- Configure automatic glossary assignment
- Configure Axon access to other systems
- Configure change request options
- Configure default values for mandatory change requests
- Configure default facets
- Configure dropdown values
- Configure email delivery options
- Configure regional settings
- Configure threshold for map objects
- Create default workflows for facet objects
- Create or edit roles and permissions
- Delete Axon objects
- Edit the glossary rules engine
- Enable fuzzy search
- Hide non-public objects from Unison search
- Enable data migration
- Enable or disable the dashboard
- Hide fields with empty values
- Import data
- Specify prefix for object reference numbers
- Specify support email address
- Use the static page editor
- View content of log files
- Transfer the ownership of dashboards and shared searches

User Profiles in Axon Data Marketplace

The following types of users can publish data, access data, and perform actions in the Data Marketplace:

Data Consumer

The data consumer is an Axon user who wants to use a data collection that is published to the Data Marketplace by a data owner. The data consumer must find the data collection and request access from the data owner. Any user in Axon Data Marketplace can be a data consumer.

For example, if a manager in an organization requests access to all the employee hire date data set, the manager is a data consumer.

Data Owner

When a data set stakeholder publishes the data sets from Axon Data Governance to the Data Marketplace, all the stakeholders for the data sets become the data owners for the data collections. By publishing the data, the stakeholder makes the data available for other Axon users who might need the data to make business decisions. Typically, the data owner is a data steward who onboards data sets to Axon, manages the data assets inventory, and defines the quality of the data asset.

For example, if the data steward of the HR team in an organization publishes the employee hire date data sets to the Data Marketplace, the data steward is a data owner.

Technical Owner for a Data Collection

The technical owner is an Axon user who has the technical rights to a data collection. The technical owner is responsible for delivering the data sets of a data collection to a data consumer.

For example, if the data consumer wants access to a database, the technical owner might be a database administrator who can grant secure access to the database. If the data consumer wants access to a spreadsheet, the technical owner might send the spreadsheet file to the data consumer by email.

Depending on the access request, the technical owner might be the same user as the data owner.

Modify a data collection to specify the technical owner for the collection. For more information, see the *Edit Data Collections* topic in the *Axon Data Governance Data Marketplace User Guide*.

Default Technical Owner for the Data Marketplace

The default technical owner is an Axon user who has the technical rights to all data collection. The technical owner is responsible for delivering the data sets of any data collection to a data consumer.

Go to the Admin Panel to assign default technical owners for all data collections. For more information, see the *Enable Data Marketplace and Assign Users* topic in the *Axon Data Governance Data Marketplace Administrator Guide*.

Data Marketplace Administrator

The Data Marketplace Administrator is the owner of all data collections and can additionally perform administrative tasks.

Go to the Admin Panel to assign Axon users with the Administrator profile in the Data Marketplace. For more information, see the *Enable Data Marketplace and Assign Users* topic in the *Axon Data Governance Data Marketplace Administrator Guide*.

SuperAdmin

All SuperAdmin users in Axon Data Governance are also SuperAdmin users in the Data Marketplace.

To see the list of tasks that are associated with the user profiles, see the *User Profiles and Tasks* topic in the *Axon Data Governance Data Marketplace User Guide*.

CHAPTER 2

Before You Run the Installer

This chapter includes the following topics:

- System Requirements, 14
- Tasks Before You Run the Installer, 24

System Requirements

Verify that the system meets the minimum hardware and software requirements for installing and running Axon.

Sizing Requirements

The sizing requirements for Axon depend on the volume of data in the facets, complexity of lineages, and concurrent users. Based on these parameters, the following levels are defined:

Low

In low-volume data, the total number of objects in a search result is less than 100,000. The total number of Policy X Attributes, Process X Attributes, and Project X Attributes links is within 5000.

Medium

In medium-volume data, the total number of objects in a seach result is between 100,000 and 200,000. The total number of Policy X Attributes, Process X Attributes, and Project X Attributes links is within 10000.

High

In high-volume data, the total number of objects in a search result is more than 200,000. The total number of Policy X Attributes, Process X Attributes, and Project X Attributes links is more than 10000. You can have a total of one million objects in Axon.

If your requirements are larger than high-level volume, contact Informatica Global Customer Support.

Based on the appropriate complexity level, you need to add resources with sufficient size and tune parameters for optimum performance.

Minimum Hardware Requirements

The following table contains the minimum system requirements for installing and running up to 10 concurrent users:

Volume	CPU Cores	Available Memory	Disk Space
Low	8 cores	32 GB	100 GB
Medium	16 cores	40 GB	100 GB
High	24 cores	48 GB	100 GB

If there is an increase in volume of data, complexity of lineages, or user concurrency, contact Informatica Global Customer Support to determine the appropriate hardware allocation.

The following table indicates the minimum system requirements for installing and running up to 10 concurrent users when you deploy the PostgreSQL database on an external node:

Parameter	Minimum Requirements
CPU Cores	2 cores
Available Memory	2 GB
Disk Space	100 GB

Recommended Hardware Specifications for Optimum Performance

Informatica recommends that you use Intel CPUs for Axon. The following table contains the hardware specifications that are recommended for optimal performance of Axon:

Component	Metric	Specification
CPU	Clock Speed	2.8 - 3.2 GHz
Memory	Frequency	DDR4
Disk	Type	SSD or SATA/SCSI. If you use the SATA/SCSI disk, ensure that the cached read speed is 10 GB/s and the cached write speed is 1.8 GB/s.
Network	Latency	< 0.2 ms

You can neither install Axon on the same server where Enterprise Data Catalog is installed, nor install it on the same server where the Hadoop cluster for Enterprise Data Catalog is installed.

For the Elasticsearch service, ensure that 15% of free disk space is available to store the records.

Important: Informatica recommends that you use a dedicated system to deploy Axon.

Software Requirements

The following table lists the minimum software requirements to run Axon:

Component	Description
Operating System	Install Axon on one of the following Linux distributions: - Amazon Linux version 2023.6 - Oracle Linux Enterprise version 8.0 to 8.10 - Oracle Linux Enterprise version 9.0 to 9.7 - Red Hat Enterprise Linux version 8.x - Red Hat Enterprise Linux version 9.0 to 9.8 - Red Hat Enterprise Linux version 10.0 to 10.1 - Rocky Linux version 9.2 to 9.6 - SUSE Linux Enterprise Server version 15 SP4, SP5, SP6, or SP7
Applications	Axon requires the following application packages: - unzip - tar - RPM - ln - setfacl - yum - zypper/rpm. Required for SUSE Linux Enterprise Server. - sed - sshpass. Required if you plan to deploy the graph database cluster or the PostgreSQL database on external nodes. **Note:** Make sure that sshpass version is less than 1.10. - Python

Port Requirements

The packages and services that Axon installs use unique ports. You can choose to change some of the port numbers during installation. The installer sets up the ports for some of the third party services and designates a range of dynamic ports to use for some application services. Verify that the port numbers are available before you start the installation process.

Axon uses the internal ports for internal communication, and the external ports for external communication.

The following table describes the default ports used by the services:

Port Type	Default Port Number	Internal or External Port
Authorization service port	7621	Internal
Bulk Upload service port	7000	Internal
Bulk Upload service pre-validator port	7010	Internal
Bulk Upload validator port	7020	Internal
Bulk Upload committer port	7030	Internal
Change Request service port	5000	Internal

Port Type	Default Port Number	Internal or External Port
Data Marketplace	7704	Internal
Document Upload service port	4000	Internal
HTTP port	9999	External
HTTPS port for SSL communication	9443	External
Memcached port	11211	Internal
Notification service port	3000	Internal
Phone Home port **Note:** Keep this port free only if you want to use the Phone Home feature. For information on the Phone Home feature, see the *Send Diagnostic Information to Informatica Global Customer Support* topic in the *Axon Data Governance 7.3.8 Administrator Guide*.	7777	Internal
Unison Search service port	7060	Internal
Value List service port	7050	Internal
Workflow service port	8081	Internal

The following table describes the port used by PostgreSQL database:

Port Type	Default Port Number	Internal or External Port
PostgreSQL Database Port **Note:** Make sure this port number is available exclusively for the PostgreSQL database.	5432	External

The following table describes the ports used by the process map configuration:

Port Type	Default Port Number	Internal or External Port
Graph Database HTTP Port	2480	Internal
Graph Database Internal Port	2424	Internal

The following table describes the ports used by the message broker:

Port Type	Default Port Number	Internal or External Port
Message Broker Port with HTTP	5672	Internal
Message Broker Port with HTTPS	5671	Internal

Note: Informatica recommends that you keep the external ports open in your organization's firewall.

For the following services, make sure that the subsequent port is free:

- Authorization
- Bulk Upload
- Change Request
- Data Marketplace
- Document Upload
- Graph Database
- Message Broker
- Notification
- Unison Search
- Value List
- Workflow

Axon uses the subsequent ports for to verify the health of the services. For example, if you configure port 7000 for the Bulk Upload service, make sure that port 7001 is free for the Bulk Upload service health check.

RPM Package Manager Files

Prepare the system before you run the Axon installer. Verify that the RPM Package Manager files (RPMs) required by Axon are available for use. Contact your IT department if you need help to have these RPMs installed.

Oracle Linux 8.x, 9.x

If you install Axon on Oracle Linux version 8.x or 9.x, ensure that the following RPM packages are available in your Linux system before you run the Axon installer:

- fontconfig
- freetype
- gettext
- libjpeg-turbo
- libpng
- libXext
- libXrender
- libxslt

- libpng15
- libpng16
- libXpm
- dejavu-sans-fonts
- jq
- compat-openssl10

  Note: This is only applicable for Oracle Linux version 8.x.
- libnsl

  Note: This is only applicable for Oracle Linux version 9.x.
- openldap-compat

  Note: This is only applicable for Oracle Linux version 9.x.
- openldap-devel

  Note: This is only applicable for Oracle Linux version 9.x.
- libzip

  Note: This is only applicable for Oracle Linux version 9.x.

Note: Oracle Linux 8.x supports OpenSSL 1.1.1k. Oracle Linux 9.x supports OpenSSL 3.2.2.

Also, ensure that the following internal RPMs or higher versions are available before you run the Axon installer:

- apr-1.4.8
- apr-util-1.5.2
- glibc-2.17
- libevent-2.0.21
- libselinux-2.5
- mailcap-2.1.41
- openssl-libs-1.0.2k
- systemd-219
- yajl-2.0.4

Optionally, you might need the following RPM packages:

- Install libmcrypt, libt1, and libXpm to resolve warning messages when you run PHP commands in the Linux environment.
- Install sshpass if you plan to deploy the PostgreSQL database on an external node.

Red Hat Enterprise Linux version 8.x, 9.x, 10.x

If you install Axon on Red Hat Enterprise Linux version 8.x or 9.x or 10.x, ensure that the following RPM packages are available in your Linux system before you run the Axon installer:

- fontconfig
- freetype
- gettext
- libjpeg-turbo

- libpng
- libXext
- libXrender
- libxslt
- libpng15
- libpng16
- libXpm
- dejavu-sans-fonts
- jq
- compat-openssl10

  Note: This is only applicable for Red Hat Enterprise Linux version 8.x.
- libnsl

  Note: This is only applicable for Red Hat Enterprise Linux version 9.x.
- openldap-compat

  Note: This is only applicable for Red Hat Enterprise Linux version 9.x.
- libzip

  Note: This is only applicable for Red Hat Enterprise Linux version 9.x.

Note: Red Hat Enterprise Linux version 8.x supports OpenSSL 1.1.1k. Red Hat Enterprise Linux version 9.x supports OpenSSL 3.2.2.

Also, ensure that the following internal RPMs or higher versions are available before you run the Axon installer:

- apr-1.4.8
- apr-util-1.5.2
- glibc-2.17
- libevent-2.0.21
- libselinux-2.5
- mailcap-2.1.41
- openssl-libs-1.0.2k
- openssl-3.0.1

  Note: This is only applicable for Red Hat Enterprise Linux version 9.x.
- systemd-219

Optionally, you might need the following RPM packages:

- Install libmcrypt, libt1, and libXpm to resolve warning messages when you run PHP commands in the Linux environment.
- Install sshpass if you plan to deploy the PostgreSQL database on an external node.

Rocky Linux version 9.2, 9.3, 9.4, 9.5, or 9.6

If you install Axon on Rocky Linux version 9.2, 9.3, 9.4, 9.5, or 9.6, ensure that the following RPM packages are available in your Linux system before you run the Axon installer:

- fontconfig

- freetype
- gettext
- libjpeg-turbo
- libpng
- libXext
- libXrender
- libxslt
- libpng15
- libpng16
- libnsl
- dejavu-sans-fonts
- jq
- libXpm
- openldap-compat
- libzip

Note: Rocky Linux version 9.x supports OpenSSL 3.2.2.

Also, ensure that the following internal RPMs or higher versions are available before you run the Axon installer:

- apr-1.4.8
- apr-util-1.5.2
- glibc-2.17
- libevent-2.0.21
- libselinux-2.5
- mailcap-2.1.41
- openssl-libs-1.0.2k
- systemd-219
- yajl-2.0.4

Optionally, you might need the following RPM packages:

- Install libmcrypt, libt1, and libXpm to resolve warning messages when you run PHP commands in the Linux environment.

SUSE Linux Enterprise Server 15 SP4, SP5, SP6, or SP7

If you install Axon on SUSE Linux Enterprise Server 15 SP4, SP5, SP6, or SP7, ensure that the following RPM packages are available in your Linux system before you run the Axon installer:

- freetype
- gcc-4.8
- gettext-runtime
- libjpeg62
- libpng15-15

- jq
- libXext6
- libXrender1
- libxslt1
- libXpm4
- libonig4
- libopenssl1_0_0
- libwebp
- libpng16-16
- libzip5

  Note: This is only applicable for SUSE Linux Enterprise Server 15 SP6 and SP7.

Note: SUSE Linux Enterprise Server versions 15 SP4, and SP5 support OpenSSL 1.1.1l. SUSE Linux Enterprise Server 15 SP6 supports OpenSSL 3.1.4, and SUSE Linux Enterprise Server 15 SP7 supports OpenSSL 3.2.3.

Also ensure that the following internal RPMs or higher versions are available before you run the Axon installer:

- glibc-2.22
- libapr1-1.5.1
- libapr-util1-1.5.3
- libevent-2_0-5-2.0.21
- libselinux1-2.5
- libyajl2-2.0.1
- libjpeg62-8
- libopenssl1_0_0-1_1
- gcc4.8-7
- libwebp
- metamail-2.7.19
- openssl-1.0.2j
- systemd-sysvinit-228

Optionally, you might need the following RPM packages:

- Install libmcrypt, libt1, and libXpm to resolve warning messages when you run PHP commands in the Linux environment.
- Install sshpass if you plan to deploy the PostgreSQL database on an external node.

Amazon Linux 2023.6

If you install Axon on Amazon Linux 2023.6, ensure that the following RPM packages are available in your Linux system before you run the Axon installer:

- fontconfig
- freetype
- gettext
- libcurl

- libicu
- libjpeg-turbo
- libpng
- libwebp
- libXext
- libXrender
- libxcrypt-compat
- libxslt
- libXpm
- dejavu-sans-fonts
- jq
- libidn
- ncurses-compat-libs.x86_64
- openldap-compat
- pcre
- libzip

Note: Amazon Linux version 2023.6.20250303 supports OpenSSL 3.2.2.

Also ensure that the following internal RPMs or higher versions are available before you run the Axon installer:

- apr-1.4.8
- apr-util-1.5.2
- glibc-2.17
- libevent-2.0.21
- libselinux-2.5
- mailcap-2.1.41
- openssl-libs-1.0.2k
- systemd-219
- yajl-2.0.4

Optionally, you might need the following RPM packages:

- Install libmcrypt, libt1, and libXpm to resolve warning messages when you run PHP commands in the Linux environment.
- Install sshpass if you plan to deploy the PostgreSQL database on an external node.

Permissions for Running RPM Commands

Before you install Axon, you must provide the Axon Administrator with the required permissions to run RPM commands.

To know more about the permissions, refer to "Provide Permissions for Running RPM Commands" on page 25.

Update Cryptographic Policy

Before you install Axon version 7.3.8 on a Red Hat Enterprise Linux (RHEL) version 9.x instance with Federal Information Processing Standards (FIPS) mode enabled and OpenSSL 3.x, complete the following prerequisites:

1. To change the system-wide cryptographic policy, run the following command in a Linux terminal as a root user with Administrator privileges:
   `update-crypto-policies --set FIPS:NO-ENFORCE-EMS`
2. Restart Axon to apply the changes.

Tasks Before You Run the Installer

Perform the pre-installation tasks before you run the Axon installer. Contact the IT department of your organization if any of these tasks requires additional resources.

Create an Axon Administrator User

The root user in Linux must create a non-root user. The non-root user installs and manages Axon.

1. Create a non-root user in the Linux environment.

   Note: This user is a non-root user that must have read and write permissions for the directory where you install Axon. Do not enter the following names for the non-root user: `axon`, `axon_bulkupload`, `axon_changerequest`, `axon_data`, `axon_docupload`, `axon_notification`, `camunda`, `postgres`, `unison`.
2. Create a home directory to install Axon. Provide the non-root user with read and write permissions on the home directory.

Note: The non-root user is called the **Axon Administrator** in this guide. All subsequent tasks in Linux must be done by the Axon Administrator inside the home directory.

Create Directories

Before you run the installer file, create the following directories within the home directory of the Axon Administrator. Make sure that the Axon Administrator has read and write permissions on the home directory.

1. Create an installation directory. The installation directory is referred to as `<INSTALLATION_DIRECTORY>` in this guide.

   When you run the Axon installer file, the installer creates an `axonhome` directory within the installation directory. The installer file then installs Axon in the `<INSTALLATION_DIRECTORY>/axonhome` directory.

   Note: The PostgreSQL database service and dependent services do not start up if the socket length exceeds 104 characters. This issue occurs because the PostgreSQL database attempts to create sockets using the entire Axon directory path during installation. To avoid this issue, restrict the length of the Axon directory path that you provide during installation to 45 characters.
2. Create a temporary directory for the installer binaries and extract the installer files to this directory. This temporary directory is referred to as `<SOURCE_BINARIES_DIRECTORY>` in this guide.

   Make sure that you download the installer zip file to a local directory or a shared network drive that is mapped on your machine. You can then securely copy the installer file to this temporary directory and extract the installer files. The installer binaries contain the core Axon components and services that are installed by the installer file.

3. If you plan to integrate Axon with Informatica Data Quality, create a directory for the Axon installer to install the Axon Agent.

Provide Permissions for Running RPM Commands

Make sure that the Axon Administrator has the required permissions to execute RPM commands.

Run the following command to provide execute permission recursively for the RPM library directory:

```
chmod -R 755 /var/lib/rpm
```

Provide Permissions to the Objects Database

Make sure that the Axon Administrator has read and write permissions to the objects database. Axon uses the PostgreSQL database to store objects.

Configure Port Access

Configure your firewall to allow the Axon installer and the application that it installs to access the external ports. Optionally, disable the firewall.

Verify Host Name and IP Address Configuration

Verify that your host name has been configured with the correct IP address.

Run the following command to show the IP address of your host name:

```
hostname -i
```

Run the following command to show the fully qualified domain name of your host:

```
hostname -f
```

Configure Host Names for Related Components

In the Linux system where you want to install Axon, you must map the host names and IP addresses of machines where other related components are installed, such as the Axon Agent and the Informatica Domain.

Perform the following tasks in the Linux system where you want to install Axon:

1. Open the **hosts** file from the following location:

   ```
   /etc/hosts
   ```

2. At the end of the **hosts** file, add the IP addresses and domain name addresses for the following components:
   - Axon Agent
   - Informatica Domain
   - Enterprise Data Catalog system
   - Data Privacy Management system
3. Save the file.

Configure Clusters for Graph Database

If you want to configure the graph database in a cluster on an external instance, verify the following prerequisites:

- The nodes must be connected to each other in a cluster.
- All the nodes in a cluster must have the same non-root user for installation.
- The password must be the same for all the nodes in a cluster.
- If you want to use the HTTPS authentication, the user must have the sudo permission.

Configure ulimit Values

Axon uses a large number of files and runs several processes. To prevent errors that result from the large number of files and process, change your Linux system settings with the ulimit command.

1. Open the `/etc/security/limits.conf` file.
2. Configure the following ulimit properties:

Property	Description
nofile	Set the file descriptor limit per process to 16000 or higher. Informatica recommends 32000 file descriptors per process.
nproc	Set the user process to 1024 or higher. Informatica recommends 4096 user processes.

3. Exit and log in to the terminal for the changes to take effect.

Install Language Fonts

To use the Axon interface in a language that requires multibyte character fonts, make sure that the language fonts are installed in the Linux system where you plan to install Axon. Download and install the language fonts at the root level using the sudo user credentials.

Verify the Axon License Key

Before you deploy Axon, verify that you have the new license key available. You supply the license key value in the License Key parameter when you configure the Axon deployment. When you download the installation files from the Informatica Electronic Software Download (ESD) site, the license key is in an email message from Informatica. Copy the license key file to a directory accessible to the user account that installs the product. Contact Informatica Global Customer Support if you do not have a new license key to use with Axon.

Download Compatible Versions of Elasticsearch and Logstash Files

Informatica does not include Elasticsearch and Logstash files into its installation package. Before you install Axon, download the Elasticsearch and Logstash supported file versions 9.3.2 or later from the Elasticsearch and Logstash official websites.

Note: If you do not use the compatible versions, you cannot proceed with the installation. When you install Axon, specify the absolute path of the downloaded Elasticsearch and Logstash files. Ensure you have access to the directory used for downloading the Elasticsearch and Logstash files.

Verify License for Data Quality Rule Automation

If you choose to automate data quality rules, you must have the 'PowerExchange for JDBC' license in the Data Integration Service of your Informatica Data Quality server. This license is required because during data quality rule automation, Axon creates a connection with Informatica Data Quality of the JDBC type.

To verify whether you have the 'PowerExchange for JDBC' license, perform the following steps.

1. Go to the Informatica Data Quality administrator console, and select the license file used by the Data Integration Service.
2. Go to the **Options** tab, and in the **Connections** section verify if 'PowerExchange for JDBC' is present.

If you do not have the license, contact Informatica Global Customer Support. To apply the license, read the following Knowledge Base article: https://knowledge.informatica.com/s/article/152991

Extract the Installer Files

Before you install Axon, you must first extract the contents of the installer file to the `<SOURCE_BINARIES_DIRECTORY>` directory. This is a temporary location that holds the files that are used to install Axon.

The installer files are compressed and distributed as a .tar file. Use an extraction utility to extract the installer files to the `<SOURCE_BINARIES_DIRECTORY>` directory on your machine.

Run the Validation Utility

Informatica provides a validation utility that you can run to verify if your system meets the requirements to run the Axon installer.

Perform the following steps to run the validation utility:

1. Download the validation utility from the following URL:
   https://download.informatica.com/infa/downloads/Axon/installer-prerequisite-tester_7.3.8.zip

   Note: Use the customer login credentials provided by Informatica. Contact Informatica Global Customer Support for assistance.
2. Copy the `installer-prerequisite-tester_7.3.8.zip` file to your Linux machine.
3. Provide the Linux user with execute permission on the `installer-prerequisite-tester_7.3.8.zip` file.
4. Run the following command to extract the contents of the `installer-prerequisite-tester_7.3.8.zip` file:

   ```
   unzip installer-prerequisite-tester_7.3.8.zip
   ```
5. From the `installer-prerequisite-tester` folder, run the following command to execute the validation script file:

   ```
   sh validate.sh
   ```

The validation utility checks if your system meets the requirements to run Axon. The utility also checks if the internal RPMs required by Axon are present in your Linux system.

The validation utility displays status messages when it runs. Follow the instructions given in the messages before you proceed with running the Axon installer. After you run the validation utility, you can also verify the log directory for log messages.

You can run the following command to check the existing package version that is installed in your Linux system:

```
rpm -qa | grep <package_name>
```

For a list of RPMs required by the Axon installer, refer to "RPM Package Manager Files " on page 18.

Obtain Domain Details

Axon can connect to other systems to retrieve and display data. If you want to integrate Axon with these systems, you must obtain their domain details. During installation, you must enter these details for Axon to connect to these systems.

Informatica Data Quality

Axon can connect to the Informatica domain to extract profile and scorecard information from Data Quality. You must provide the following details of your Informatica domain if you choose to install the Axon Agent when you run the Axon installer:

- Domain name
- Node port number
- Model Repository Service node name
- Model Repository Service domain host name
- Model Repository Service username and password
- Model Repository Service name
- Model Repository Service security domain
- Model Repository Service SSL details
- Data Integration Service node name
- Data Integration Service domain host name
- Data Integration Service username and password
- Data Integration Service name
- Data Integration Service security domain
- Data Integration Service SSL details

Enterprise Data Catalog

Axon can discover and retrieve enterprise metadata from Enterprise Data Catalog. To integrate Axon with Enterprise Data Catalog, you must provide the following details of your Enterprise Data Catalog installation:

- Enterprise Data Catalog server URL
- Port number
- User name and password
- User security domain
- Axon Resource name

Data Privacy Management

Axon can display the security status of your data from Data Privacy Management. To integrate Axon with Data Privacy Management, you must provide the following details of your Data Privacy Management installation:

- Data Privacy Management server URL
- Port number
- User name and password
- User security domain

Note: If you plan to integrate Axon with Data Privacy Management, you must ensure that Data Privacy Management and Enterprise Data Catalog have been configured properly and are running.

Obtain SSL Details

Axon uses HTTP or HTTPS for communication. To use HTTPS communication, you must configure the Secure Sockets Layer (SSL) protocol during installation. Keep the SSL certificate and key file ready before you start the installation.

You must have the following information ready before you start the installation:

- HTTPS port information
- SSL certificate and the path to the file. The certificate file must be in the CRT or PEM format.
- SSL key file and the path to the file. The key must be in .key format.

Copy the SSL certificate file and SSL key file to a directory on which the Axon Administrator has read and write permissions.

Note: If the SSL certificate is in the DER format, go to the certificate location and run the following command to convert the certificate to the PEM format:

```
openssl x509 -in <DER_file_name> -inform der -outform pem -out <PEM_file_output_name>
```

For example, if the SSL certificate file is `cert.der`, and you want to convert the file to `cert.pem`, run the following command:

```
openssl x509 -in cert.der -inform der -outform pem -out cert.pem
```

When Axon connects to other Informatica products and acts as the server, Axon supports TLS versions 1.2 and 1.3. Make sure that the TLS version is correctly specified in the `app.tlsVersion` parameter of the `<Axon_Agent_Home>/conf/application-app.properties` file.

Create a New Database Schema to Automate Data Quality Rules

If you want to use Informatica Data Quality to automate data quality rules, verify the following prerequisites for the database where Informatica Data Quality stores the data quality scores:

- You have CREATE, EDIT, ALTER, SELECT and UPDATE privileges on the database.
- A new schema is created on the database.
- The database is a relational database.
- The database supports native connections.
  For more information on the database systems that are supported by native connection, see the *Native Connectivity* topic in the *Informatica Data Quality Administrator Guide*.

CHAPTER 3

Installer Directory Structure

When you extract the installer file, the extraction utility decompresses the Axon installation files and places them in the `<SOURCE_BINARIES_DIRECTORY>` directory.

After you extract the installer file, you can find the following files and directories in the `<SOURCE_BINARIES_DIRECTORY>` directory:

- `install.sh` file
- `silentinstall.sh` file
- `Messages` directory
- `SilentInput_AxonAgent.properties` file
- `SilentInput.properties` file
- `Server` directory
- `properties` directory
- `source` directory
- The installer .tar file
- `log` directory

CHAPTER 4

Running the Installer

This chapter includes the following topics:

- Running the Installer Overview, 31
- Running the Installer in Console Mode, 31
- Running the Installer in Silent Mode, 47

Running the Installer Overview

Informatica provides an installer to install Axon Data Governance on a machine that runs Linux. You can run only one instance of Axon on a machine at any given point of time. If you want to enable communication between Axon and Informatica Data Quality, you can choose to install the Axon Agent when you install Axon.

You can install Axon or the Axon Agent in one of the following ways:

- Console mode. Interact with the installer wizard to configure the installation parameters. After you run the installer file, the interactive wizard prompts you to enter the parameters at each step of the installation.
- Silent mode. Install without user intervention. Enter the installation parameters in the installer properties file before you start the installation. After you have prepared the installer properties file, run the installer file. The installer reads the parameters from the properties files, runs in the background, and installs Axon or the Axon Agent.

The console mode installer and silent mode installer files are located in the `<SOURCE_BINARIES_DIRECTORY>` directory.

Running the Installer in Console Mode

Install Axon using a console. When you run the installer, the words `Quit`, `Back`, and `Help` are reserved words. Do not use them as input text. Enter `help` for getting additional context-sensitive information during the installation process. Enter `Quit` to exit the installer at any point in time.

1. Log in to the machine as the Axon Administrator.
2. Ensure that no Axon-related service is running. If you are running an earlier version of Axon, go to the `<INSTALLATION_DIR>/bin` directory and run the following command to shut down Axon and all Axon services:

```
shutdown.sh
```

3. From the command line, run the `install.sh` file.
4. Enter the number corresponding to one of the following options to continue:

Option	Description
1. Install Axon.	Run the installer to install Axon.
2. Install Axon Agent.	Run the installer to install the Axon Agent.
3. Install both Axon and Axon Agent.	Run the installer to install both Axon and the Axon Agent.

Installing Axon

Select the option to install only Axon if you do not want to install the Axon Agent. Informatica recommends that you run the validation utility before you run the Axon installer.

Run the `install.sh` file, and select option 1 **Install only Axon**.

The installer validates at every step. If you enter a wrong value, the installation stops.

Note: When you install Axon, the installer prompts you to enter the port numbers of the Axon services. The installer checks if the ports you entered are free or already used. At each prompt, press the **Enter** key to proceed with the default port. If you want to enter custom port numbers, make sure that the port numbers are in the range of 1025 to 65535.

Step 1 of 8

Review the information on the screen, and type **1** to continue with the installation.

To upgrade your Axon installation to a newer version, type **2**. For more information, see the *Axon Data Governance Upgrade Guide* in the Informatica Documentation Portal.

Step 2 of 8

Enter the following installation details:

- Directory where you want to install Axon. Press the **Enter** key to proceed with the default directory.
- Fully qualified file path of the new or updated license key.

  Note: If you install Axon as the Axon Administrator, you must have read permission for the directory where the license key file is located.
- Fully qualified path of the Elasticsearch file.
- Fully qualified path of the Logstash file.
- Path to your cryptographic private key. Press the **Enter** key to proceed with the default key that the Axon installer provides. The installer uses this key to encrypt the passwords that you type during installation.

  Note: The private key file must have a minimum of 16 characters.
- Option to connect to the Axon server using a proxy host name. If you do not select this option, the installer uses the URL of the Axon server as the host name. If you select this option, enter the proxy host name.
  The installer performs a check to validate if you have installed Python. Run the which Python command to retrieve the Python path.

- Option to allow the Axon SuperAdmin to view log files in the Axon Admin Panel. If you enable this option now, you can disable it any time after installation is complete.

  Note: If you enable this option, make sure that the Linux system has 32 GB of memory and 10 CPU cores. For more information on the requirements to run Axon, refer to "System Requirements" on page 14.

The installer performs a check to validate the system requirements and verifies whether the license key and cryptographic private key files exist in the directory locations that you have provided.

Step 3 of 8

Axon Data Governance uses a service called Event Monitor that collects usage data. You can use the data to gain insight into how your organization uses Axon. Review this step to verify whether you want to enable it during the installation or not. If you do not enable this service during the installation, you can enable it after the installation is completed. After you enable the service, you cannot disable it.

Step 4 of 8

If you install Axon as a root user, specify the Axon Administrator that installs and manages Axon. This user is a non-root user that must have read and write permission on the Axon installation directory.

Step 5 of 8

Review the pre-installation summary information. Press the **Enter** key to continue. If you want to make changes to the information you provided, type in **Back** and press **Enter**.

Step 6 of 8

Review the installation progress.

Step 7 of 8

After the installation progress reaches 100%, you can enable HTTPS authentication, choose different deployment options for the PostgreSQL database, and configure various advanced settings.

In the advanced settings, you can configure the following options:

- Configure HTTPS authentication.
- You can choose to configure the PostgreSQL database on a different node, external PostgreSQL database instances, such as Amazon RDS and Microsoft Azure Database, or use the default PostgreSQL deployment.

  Note: Effective in version 7.3.3 or later, the Axon installer dropped support for Microsoft Azure PostgreSQL Single Server. You can instead use the Microsoft Azure PostgreSQL Flexible Server.
- You can customize database user passwords for the PostgreSQL databases that are used by the Axon services.
- You can configure the graph database in a cluster on external nodes for better search concurrence and performance.
- Enter custom port numbers for the services that Axon uses.

  Note:

  - Do not modify the port for the message broker service.
  - Enter a value greater than 1024 for the HTTP or HTTPS port.
  - If you enter a custom port for the following services, make sure that the subsequent port is free:
    - Authorization

- Bulk Upload
- Change Request
- Data Marketplace
- Document Upload
- Graph Database
- Message Broker
- Notification
- Unison Search
- Value List
- Workflow

Axon uses the subsequent port for each service to verify the health of the service. For example, if you configure port 7525 for the Bulk Upload service, make sure that port 7526 is free for the Bulk Upload service health check.

- Integrate Axon with Enterprise Data Catalog.
- Integrate Axon with Data Privacy Management.

Type **1** to continue with the advanced configuration. Type **2** to skip to step 6.

Step 7A of 8

Type **1** if you want to enable HTTPS authentication, and then enter the values for the following parameters:

- HTTPS port number
- File path to the SSL certificate. The certificate file must be in CRT or PEM format. If your SSL certificate is protected by a password, the HTTPD service cannot start. You are prompted for the password each time the HTTPD service is started or restarted. For more information, refer to "Cannot Start the HTTPD Service because SSL is Password-Protected" on page 67
- File path to the SSL key. The key must be in `.key` format.
- Memcached port number

Type **2** if you do not want to enable HTTPS authentication.

Select the mode of deployment for the PostgreSQL database. If you choose to use the default PostgreSQL deployment, type **3** and continue with the rest of the installation process. If you choose to deploy the PostgreSQL database on a different node, type **1** and specify the host name, user name same as that of the Axon administrator, password, and directory of the node where the PostgreSQL database must be deployed.

If you choose to deploy the PostgreSQL database on external nodes, such as Amazon RDS or Azure PostgreSQL Database, type **2** and select the external deployment platform. The Linux user that manages the PostgreSQL database must have read, write, modify and delete access on the PostgreSQL database. Additionally, the Linux user must be the same non-root user who installs and manages Axon.

If you want to deploy the PostgreSQL database on the Amazon RDS instance, type **1** and press the **Enter** key. If you want to provide a connect string for the Amazon RDS instance, type **1** and specify the connect string in the following format:

```
postgresql://db_user:db_password@hostname:db_port/db_name
```

If you do not want to provide a connect string for the Amazon RDS instance, type **2** and press the **Enter** key. Specify the host name, port number, administrator user name, administrator user password, and the existing PostgreSQL database name on the Amazon RDS instance.

If you want to deploy the PostgreSQL database on the Microsoft Azure Database instance, type **2** and press the **Enter** key. If you want to provide a connect string for the Microsoft Azure Database instance, type **1** and specify the connect string in the following format:

```
postgresql://db_user:db_password@hostname:db_port/db_name?sslmode=require
```

If you do not want to provide a connect string for the Microsoft Azure Database instance, type **2** and press the **Enter** key. Specify the host name, port number, administrator user name, administrator user password, and the existing PostgreSQL database name on the Microsoft Azure Database instance.

You can choose to configure custom password for PostgreSQL databases. If you want to configure custom passwords, type **1**. Enter the password for the `axon` database and press the **Enter** key. If you want to configure the `axon` database password for other databases, then type **1**. If you want to configure separate passwords for other databases, type **2**. Press the **Enter** key to process with providing passwords for the following databases:

- `axon`
- `axon_data`
- `axon_changerequest`
- `axon_notification`
- `axon_docupload`
- `axon_bulkupload`
- `unison`
- `marketplace`
- `segmentation`

Note:

1. If you choose to provide custom passwords, ensure that you save the passwords in a file for future uses, such as for content migration.
2. You cannot provide custom password for the `camunda` database during installation because the Workflow service cannot read an encrypted password. You can customize the password for the `camunda` database after you install Axon.

If you want to configure an external graph database cluster and you have not enabled the HTTPS authentication, you need to enter the following details:

- Number of nodes in the cluster. Informatica recommends that you add a minimum of three nodes for a graph database cluster.
- Host names of all the nodes in the cluster. You have to enter the host names separated by commas.
- User name and password for the cluster. You have to enter a single user name password for all the nodes in a cluster.
- Directory where the graph database cluster has to be deployed.

If you want to configure an external graph database cluster and you have enabled the HTTPS authentication, you need to enter the following details:

- Number of nodes in the cluster. Informatica recommends that you add a minimum of three nodes for a graph database cluster.
- Host names of all the nodes in the cluster. You have to enter each host name separated by a comma. Ensure that the DNS entries contain these host names, or you need to add the mapping in the `/etc/hosts` file.

- Sudo user name and password for the cluster. You have to enter a single user name with sudo user privilege and password for all the nodes in a cluster.
- Directory where the graph database cluster has to be deployed.
- Private Virtual Private Network (VPN) IP addresses for the nodes where you want to deploy the graph database cluster. Provide each IP address separated by a comma. The IP addresses are required to create a secure VPN using Nebula. The secured VPN encrypts traffic between the graph database nodes in the cluster.

Step 7B of 8

Specify the port numbers of the following services. At each step, press the **Enter** key to proceed with the default port.

- Workflow service port
- Document upload port
- Change request service port
- Bulk upload service port
- Bulk upload pre-validator port
- Bulk upload validator port
- Bulk upload committer port
- Notification service port
- Unison search service port
- Value list service port

Step 7C of 8

Press the **Enter** key to proceed with the default port numbers for the OrientDB database configurations.

- Process configuration HTTP port
- Process configuration internal port

Note: Do not enter custom port numbers for the OrientDB database configurations.

Step 7D of 8

Specify the port number for the message broker. Press the **Enter** key to proceed with the default port.

Step 7E of 8

Type **1** to integrate Axon with Enterprise Data Catalog. Type **2** to skip to step 7.

Note: If you want to integrate Axon with Enterprise Data Catalog at a later stage after installation, refer to "Configure Access to Enterprise Data Catalog" on page 49.

Type **1** to integrate Axon with Data Privacy Management. Type **2** to skip to step 7.

Note: If you want to integrate Axon with Data Privacy Management at a later stage after installation, refer to "Configure Access to Data Privacy Management" on page 52.

Step 7F of 8

Enter values for the following parameters of the Enterprise Data Catalog installation:

- Enterprise Data Catalog host name in the following format: `http(s)://<host_name>`
- Enterprise Data Catalog port number. Press the **Enter** key to proceed with the default port.
- Enterprise Data Catalog user name

- Enterprise Data Catalog password
- Enterprise Data Catalog user security domain
- Name of the Axon resource created in Enterprise Data Catalog that connects to your Axon application

Step 7G of 8

Enter values for the following parameters of the Data Privacy Management installation:

- Data Privacy Management host name in the following format: `http(s)://<host_name>`
- Data Privacy Management port number. Press the **Enter** key to proceed with the default port.
- Data Privacy Management user name
- Data Privacy Management password
- Data Privacy Management user security domain

The installer begins installing Axon.

Note: The Axon installation process might take some time to complete. To view the status of the installation, open the file `<INSTALLATION_DIRECTORY>/Configure.log`.

Step 8 of 8

Review the installation summary to verify that all components are successfully installed.

Press the **Enter** key to complete the installation.

Optionally, run the `reset` command to reset the Linux terminal.

Installing the Axon Agent

Select the option to install the Axon Agent if you want to connect Axon to Informatica Data Quality to run data quality rules on the Axon data assets. Informatica recommends that you install the Axon Agent on the system where you have installed Informatica Data Quality. You must know the details of the Informatica domain before you proceed.

Run the `install.sh` file, and select option 2 **Install only the Axon Agent**.

Note: The console is refreshed and the installation screen is cleared. You might want to note down any important information on the screen.

Step 1 of 9

Select option 1 to install the Axon Agent, and press the **Enter** key to continue.

Step 2 of 9

Enter the following installation details:

- Version of the Informatica domain that you want Axon to connect to.
- If you run the `install.sh` file as a root user, specify the non-root user that installs and manages the Axon Agent. The non-root user must have read and write permission on the Axon Agent installation directory.
- Option to automate data quality rules. If you choose this option, Axon automatically runs data quality rules on Axon attributes, generates data quality scores in Informatica Data Quality, and displays the scores in the Axon interface. For more information, refer the *Automate Data Quality Rules* chapter in the *Axon Data Governance User Guide*.
- If you choose to automate data quality rules, enter the connection name of the database where Informatica Data Quality stores the data quality scores. For more information on the pre-requisites, see "Create a New Database Schema to Automate Data Quality Rules" on page 29.

Step 3 of 9

Enter the path to the directory where you want the installer to install the Axon Agent.

Note: Install the Axon Agent in a directory other than the Axon installation directory.

Step 4 of 9

Review the pre-installation summary information. Press the **Enter** key to continue. If you want to make changes to the information you provided, type **Back** and press **Enter**.

Step 5 of 9

Review the installation progress.

Step 6 of 9

Specify the details of the Model Repository Service (MRS). Enter values for the following parameters:

- Domain name
- Node port
- MRS domain host
- MRS node name
- MRS user name
- MRS password
- MRS service name
- Security domain of the MRS user. Choose LDAP or your native domain as the security domain. Press the **Enter** key to proceed with the native domain as default.
- SSL security. Specify whether SSL security is enabled for your MRS system. Press the **Enter** key to proceed with no SSL security as the default option. If you select yes, you must provide the local path to the trust store file.

  Note: The trust store file name is usually `infa_truststore.jks`. Informatica recommends that you first copy the file from the `<Informatica_Data_Quality_Installed_Location>/services/shared/security` directory to the system where you want to install the Axon Agent, and then provide the local path to the file.
- Custom SSL authentication. Specify whether your MRS system uses custom SSL authentication. Press the **Enter** key to proceed with no custom SSL authentication as the default option. If you select yes, you must provide the SSL password.

Step 7 of 9

Specify the details of the Data Integration Service (DIS). Enter values for the following parameters:

- Node port
- DIS domain host
- DIS node name
- DIS user name
- DIS password
- DIS service name
- Security domain of the DIS user. Choose LDAP or your native domain as the security domain. Press the **Enter** key to proceed with the native domain as default.

  Note: If you choose LDAP as your security domain, you must point the Axon Agent to connect to your organization's LDAP server. For more information, see the *Modify Connection Details* topic in the *Monitoring Axon* chapter of the *Axon Data Governance Administrator Guide*.

- SSL security. Specify whether SSL security is enabled for your DIS system. Press the **Enter** key to proceed with no SSL security as the default option. If you select yes, you must provide the local path to the trust store file.

  Note: The trust store file name is usually `infa_truststore.jks`. Informatica recommends that you first copy the file from the `<Informatica_Data_Quality_Installed_Location>/services/shared/security` directory to the system where you want to install the Axon Agent, and then provide the local path to the file.
- Custom SSL authentication. Specify whether your DIS system uses custom SSL authentication. Press the **Enter** key to proceed with no custom SSL authentication as the default option. If you select yes, you must provide the SSL password.

Step 8 of 9

Specify the details of the Axon installation. The Axon Agent uses the parameters you provide here to connect to Axon. Enter values for the following parameters:

- HTTPS authentication

  Note: If you plan to automate data quality rules, connect Axon to the Axon Agent using the HTTP protocol. If you choose to automate data quality rules, Axon automatically runs data quality rules on fields scanned by Enterprise Data Catalog and displays the scores in Axon. For more information, refer the *Automate Data Quality Rules* chapter in the *Axon Data Governance User Guide*.
- Axon URL in the format `http(s)://<host_name>:<9999_or_9443_or_custom_port_number>`

  Note: The default port number for HTTP authentication is 9999, and the default port number for HTTPS authentication is 9443. Do not enter a forward slash (/) after the URL.
- Axon user name or LDAP user name to access the Axon application

  Note: If you use Single Sign-On (SSO) for Axon, do not enter the SSO user name. Enter the user name that is configured in the Axon application.
- Axon password or LDAP password to access the Axon application

  Note: If you use Single Sign-On (SSO) for Axon, do not enter the SSO password. Enter the password that is configured in the Axon application.
- HTTP port where you want to run the Axon Agent.

Step 9 of 9

Review the installation summary to verify that all components are successfully installed.

Press the **Enter** key to complete the installation.

Optionally, run the `reset` command to reset the Linux terminal.

To verify that the Axon Agent is running, open the file `<Axon_Agent_Home>/logs/dq-axon/dqagent.log`

Installing Axon and the Axon Agent

Select the option to install Axon and the Axon Agent to enable communication with Informatica Data Quality. Informatica recommends that you run the validation utility before you run the Axon installer.

Run the `install.sh` file, and select option 3 **Install both Axon and the Axon Agent**.

The installer will first install Axon. The installer validates at every step. If you enter a wrong value, the installation stops.

Note: When you install Axon, the installer prompts you to enter the port numbers of the Axon services. The installer checks if the ports you entered are free or already used. At each prompt, press the **Enter** key to

proceed with the default port. If you want to enter custom port numbers, make sure that the port numbers are in the range of 1025 to 65535.

Step 1 of 8

Review the information on the screen, and type **1** to continue with the installation.

To upgrade your Axon installation to a newer version, type **2**. For more information, see the *Axon Data Governance Upgrade Guide* in the Informatica Documentation Portal.

Step 2 of 8

Enter the following installation details:

- Directory where you want to install Axon. Press the **Enter** key to proceed with the default directory.
- Fully qualified file path of the new or updated license key.

  Note: If you install Axon as the Axon Administrator, you must have read permission for the directory where the license key file is located.
- Fully qualified path of the Elasticsearch file.
- Fully qualified path of the Logstash file.
- Path to your cryptographic private key. Press the **Enter** key to proceed with the default key that the Axon installer provides. The installer uses this key to encrypt the passwords that you type during installation.

  Note: The private key file must have a minimum of 16 characters.
- Option to connect to the Axon server using a proxy host name. If you do not select this option, the installer uses the URL of the Axon server as the host name. If you select this option, enter the proxy host name.
  The installer performs a check to validate if you have installed Python. Run the which Python command to retrieve the Python path.
- Option to allow the Axon SuperAdmin to view log files in the Axon Admin Panel. If you enable this option now, you can disable it any time after installation is complete.

  Note: If you enable this option, make sure that the Linux system has 32 GB of memory and 10 CPU cores. For more information on the requirements to run Axon, refer to "System Requirements" on page 14.

The installer performs a check to validate the system requirements and verifies whether the license key and cryptographic private key files exist in the directory locations that you have provided.

Step 3 of 8

Axon Data Governance uses a service called Event Monitor that collects usage data. You can use the data to gain insight into how your organization uses Axon. Review this step to verify whether you want to enable it during the installation or not. If you do not enable this service during the installation, you can enable it after the installation is completed. After you enable the service, you cannot disable it.

Step 4 of 8

If you install Axon as a root user, specify the Axon Administrator that installs and manages Axon. This user is a non-root user that must have read and write permission on the Axon installation directory.

Step 5 of 8

Review the pre-installation summary information. Press the **Enter** key to continue. If you want to make changes to the information you provided, type in **Back** and press **Enter**.

Step 6 of 8

Review the installation progress.

Step 7 of 8

After the installation progress reaches 100%, you can enable HTTPS authentication, choose different deployment options for the PostgreSQL database, and configure various advanced settings.

In the advanced settings, you can configure the following options:

- Configure HTTPS authentication.
- You can choose to configure the PostgreSQL database on a different node, external PostgreSQL database instances, such as Amazon RDS and Microsoft Azure Database, or use the default PostgreSQL deployment.

  Note: Effective in version 7.3.3 or later, the Axon installer dropped support for Microsoft Azure PostgreSQL Single Server. You can instead use the Microsoft Azure PostgreSQL Flexible Server.
- You can customize database user passwords for the PostgreSQL databases that are used by the Axon services.
- You can configure the graph database in a cluster on external nodes for better search concurrence and performance.
- Enter custom port numbers for the services that Axon uses.

  Note:

  - Do not modify the port for the message broker service.
  - Enter a value greater than 1024 for the HTTP or HTTPS port.
  - If you enter a custom port for the following services, make sure that the subsequent port is free:
    - Authorization
    - Bulk Upload
    - Change Request
    - Data Marketplace
    - Document Upload
    - Graph Database
    - Message Broker
    - Notification
    - Unison Search
    - Value List
    - Workflow

    Axon uses the subsequent port for each service to verify the health of the service. For example, if you configure port 7525 for the Bulk Upload service, make sure that port 7526 is free for the Bulk Upload service health check.
- Integrate Axon with Enterprise Data Catalog.
- Integrate Axon with Data Privacy Management.

Type **1** to continue with the advanced configuration. Type **2** to skip to step 6.

Step 7A of 8

Type **1** if you want to enable HTTPS authentication, and then enter the values for the following parameters:

- HTTPS port number

- File path to the SSL certificate. The certificate file must be in CRT or PEM format. If your SSL certificate is protected by a password, the HTTPD service cannot start. You are prompted for the password each time the HTTPD service is started or restarted. For more information, refer to "Cannot Start the HTTPD Service because SSL is Password-Protected" on page 67
- File path to the SSL key. The key must be in .key format.
- Memcached port number

Type **2** if you do not want to enable HTTPS authentication.

Select the mode of deployment for the PostgreSQL database. If you choose to use the default PostgreSQL deployment, type **3** and continue with the rest of the installation process. If you choose to deploy the PostgreSQL database on a different node, type **1** and specify the host name, user name same as that of the Axon administrator, password, and directory of the node where the PostgreSQL database must be deployed. The Linux user that manages the PostgreSQL database must have read, write, modify and delete access on the PostgreSQL database. Additionally, the Linux user must be the same non-root user who installs and manages Axon.

If you choose to deploy the PostgreSQL database on external nodes, such as Amazon RDS or Azure PostgreSQL Database, type **2** and select the external deployment platform.

If you want to deploy the PostgreSQL database on the Amazon RDS instance, type **1** and press the **Enter** key. If you want to provide a connect string for the Amazon RDS instance, type **1** and specify the connect string in the following format:

```
postgresql://db_user:db_password@hostname:db_port/<db_name>
```

If you do not want to provide a connect string for the Amazon RDS instance, type **2** and press the **Enter** key. Specify the host name, port number, administrator user name, administrator user password, and the existing PostgreSQL database name on the Amazon RDS instance.

If you want to deploy the PostgreSQL database on the Microsoft Azure Database instance, type **2** and press the **Enter** key. If you want to provide a connect string for the Microsoft Azure Database instance, type **1** and specify the connect string in the following format:

```
postgresql://db_user:db_password@hostname:db_port/db_name?sslmode=require
```

If you do not want to provide a connect string for the Microsoft Azure Database instance, type **2** and press the **Enter** key. Specify the host name, port number, administrator user name, administrator user password, and the existing PostgreSQL database name on the Microsoft Azure Database instance.

You can choose to configure custom password for PostgreSQL databases. If you want to configure custom passwords, type **1**. Enter the password for the `axon` database and press the **Enter** key. If you want to configure the `axon` database password for other databases, then type **1**. If you want to configure separate passwords for other databases, type **2**. Press the **Enter** key to process with providing passwords for the following databases:

- `axon`
- `axon_data`
- `axon_changerequest`
- `axon_notification`
- `axon_docupload`
- `axon_bulkupload`
- `unison`
- `marketplace`
- `segmentation`

Note:

1. If you choose to provide custom passwords, ensure that you save the passwords in a file for future uses, such as for content migration.
2. You cannot provide custom password for the `camunda` database during installation because the Workflow service cannot read an encrypted password. You can customize the password for the `camunda` database after you install Axon.

If you want to configure an external graph database cluster and you have not enabled the HTTPS authentication, you need to enter the following details:

- Number of nodes in the cluster. Informatica recommends that you add a minimum of three nodes for a graph database cluster.
- Host names of all the nodes in the cluster. You have to enter the host names separated by commas.
- User name and password for the cluster. You have to enter a single user name password for all the nodes in a cluster.
- Directory where the graph database cluster has to be deployed.

If you want to configure an external graph database cluster and you have enabled the HTTPS authentication, you need to enter the following details:

- Number of nodes in the cluster. Informatica recommends that you add a minimum of three nodes for a graph database cluster.
- Host names of all the nodes in the cluster. You have to enter each host name separated by a comma. Ensure that the DNS entries contain these host names, or you need to add the mapping in the `/etc/hosts` file.
- Sudo user name and password for the cluster. You have to enter a single user name with sudo user privilege and password.
- Directory where the graph database cluster has to be deployed.
- Private Virtual Private Network (VPN) IP addresses for the nodes where you want to deploy the graph database cluster. Provide each IP address separated by a comma. The IP addresses are required to create a secure VPN using Nebula. The secured VPN encrypts traffic between the graph database nodes in the cluster.

Step 7B of 8

Specify the port numbers of the following services. At each step, press the **Enter** key to proceed with the default port.

- Workflow service port
- Document upload port
- Change request service port
- Bulk upload service port
- Bulk upload pre-validator port
- Bulk upload validator port
- Bulk upload committer port
- Notification service port
- Unison search service port
- Value list service port

Step 7C of 8

Specify the port numbers for the OrientDB database configurations. At each step, press the **Enter** key to proceed with the default port.

- Process configuration HTTP port
- Process configuration internal port

Step 7D of 8

Specify the port number for the message broker. Press the **Enter** key to proceed with the default port.

Step 7E of 8

Type **1** to integrate Axon with Enterprise Data Catalog. Type **2** to skip to step 7.

Note: If you want to integrate Axon with Enterprise Data Catalog at a later stage after installation, refer to "Configure Access to Enterprise Data Catalog" on page 49.

Type **1** to integrate Axon with Data Privacy Management. Type **2** to skip to step 7.

Note: If you want to integrate Axon with Data Privacy Management at a later stage after installation, refer to "Configure Access to Data Privacy Management" on page 52.

Step 7F of 8

Enter values for the following parameters of the Enterprise Data Catalog installation:

- Enterprise Data Catalog host name in the following format: `http(s)://<host_name>`
- Enterprise Data Catalog port number. Press the **Enter** key to proceed with the default port.
- Enterprise Data Catalog user name
- Enterprise Data Catalog password
- Enterprise Data Catalog user security domain
- Name of the Axon resource created in Enterprise Data Catalog that connects to your Axon application

Step 7G of 8

Enter values for the following parameters of the Data Privacy Management installation:

- Data Privacy Management host name in the following format: `http(s)://<host_name>`
- Data Privacy Management port number. Press the **Enter** key to proceed with the default port.
- Data Privacy Management user name
- Data Privacy Management password
- Data Privacy Management user security domain

The installer begins installing Axon.

Note: The Axon installation process might take some time to complete. To view the status of the installation, open the file `<INSTALLATION_DIRECTORY>/Configure.log`.

Step 8 of 8

Review the installation summary to verify that all components are successfully installed.

Press the **Enter** key to complete the installation.

Continue with the installation of the Axon Agent.

Note: The console is refreshed and the installation screen is cleared. You might want to note down any important information on the screen.

Step 1 of 9

Select option 1 to install the Axon Agent, and press the **Enter** key to continue.

Step 2 of 9

Enter the following installation details:

- Version of the Informatica domain that you want Axon to connect to.
- If you run the `install.sh` file as a root user, specify the non-root user that installs and manages the Axon Agent. The non-root user must have read and write permission on the Axon Agent installation directory.
- Option to automate data quality rules. If you choose this option, Axon automatically runs data quality rules on Axon attributes, generates data quality scores in Informatica Data Quality, and displays the scores in the Axon interface. For more information, refer the *Automate Data Quality Rules* chapter in the *Axon Data Governance User Guide*.
- If you choose to automate data quality rules, enter the connection name of the database where Informatica Data Quality stores the data quality scores. For more information on the pre-requisites, see "Create a New Database Schema to Automate Data Quality Rules" on page 29.

Step 3 of 9

Enter the path to the directory where you want the installer to install the Axon Agent.

Note: Install the Axon Agent in a directory other than the Axon installation directory.

If you install the Axon Agent as a root user, specify the Axon Administrator that installs and manages Axon. This user is a non-root user that must have read and write permission on the Axon Agent installation directory.

Step 4 of 9

Review the pre-installation summary information. Press the **Enter** key to continue. If you want to make changes to the information you provided, type **Back** and press **Enter**.

Step 5 of 9

Review the installation progress.

Step 6 of 9

Specify the details of the Model Repository Service (MRS). Enter values for the following parameters:

- Domain name
- Node port
- MRS domain host
- MRS node name
- MRS user name
- MRS password
- MRS service name
- Security domain of the MRS user. Choose LDAP or your native domain as the security domain. Press the **Enter** key to proceed with the native domain as default.
- SSL security. Specify whether SSL security is enabled for your MRS system. Press the **Enter** key to proceed with no SSL security as the default option. If you select yes, you must provide the local path to the trust store file.

  Note: The trust store file name is usually `infa_truststore.jks`. Informatica recommends that you first copy the file from the `<Informatica_Data_Quality_Installed_Location>/services/shared/`

`security` directory to the system where you want to install the Axon Agent, and then provide the local path to the file.

- Custom SSL authentication. Specify whether your MRS system uses custom SSL authentication. Press the **Enter** key to proceed with no custom SSL authentication as the default option. If you select yes, you must provide the SSL password.

Step 7 of 9

Specify the details of the Data Integration Service (DIS). Enter values for the following parameters:

- Node port
- DIS domain host
- DIS node name
- DIS user name
- DIS password
- DIS service name
- Security domain of the DIS user. Choose LDAP or your native domain as the security domain. Press the **Enter** key to proceed with the native domain as default.

  Note: If you choose LDAP as your security domain, you must point the Axon Agent to connect to your organization's LDAP server. For more information, see the *Modify Connection Details* topic in the *Monitoring Axon* chapter of the *Axon Data Governance Administrator Guide*.
- SSL security. Specify whether SSL security is enabled for your DIS system. Press the **Enter** key to proceed with no SSL security as the default option. If you select yes, you must provide the local path to the trust store file.

  Note: The trust store file name is usually `infa_truststore.jks`. Informatica recommends that you first copy the file from the `<Informatica_Data_Quality_Installed_Location>/services/shared/security` directory to the system where you want to install the Axon Agent, and then provide the local path to the file.
- Custom SSL authentication. Specify whether your DIS system uses custom SSL authentication. Press the **Enter** key to proceed with no custom SSL authentication as the default option. If you select yes, you must provide the SSL password.

Step 8 of 9

Specify the details of the Axon installation. The Axon Agent uses the parameters you provide here to connect to Axon. Enter values for the following parameters:

- HTTPS authentication.

  Note: If you plan to automate data quality rules, connect Axon to the Axon Agent using the HTTP protocol. If you choose to automate data quality rules, Axon automatically runs data quality rules on fields scanned by Enterprise Data Catalog and displays the scores in Axon. For more information, refer the *Automate Data Quality Rules* chapter in the *Axon Data Governance User Guide*.
- Axon URL in the format `http(s)://<host_name>:<9999_or_9443_or_custom_port_number>`

  Note: The default port number for HTTP authentication is 9999, and the default port number for HTTPS authentication is 9443. Do not enter a forward slash (/) after the URL.
- Axon user name or LDAP user name to access the Axon application

  Note: If you use Single Sign-On (SSO) for Axon, do not enter the SSO user name. Enter the user name that is configured in the Axon application.
- Axon password or LDAP password to access the Axon application

Note: If you use Single Sign-On (SSO) for Axon, do not enter the SSO password. Enter the password that is configured in the Axon application.

- HTTP port where you want to run the Axon Agent.

Step 9 of 9

Review the installation summary to verify that all components are successfully installed.

Press the **Enter** key to complete the installation.

Optionally, run the `reset` command to reset the Linux terminal.

To verify that the Axon Agent is running, open the file `<Axon_Agent_Home>/logs/dq-axon/dqagent.log`

Running the Installer in Silent Mode

If you want to install Axon or the Axon Agent without user intervention, choose to install in silent mode. If the passwords in the `silentinstall.sh` file are stored in unencrypted form, delete the file from your Linux system after you install Axon.

1. Go to the `<SOURCE_BINARIES_DIRECTORY>` directory, and enter the Axon installation parameters in the `SilentInput.properties` file.
2. If you want to integrate Axon with Informatica Data Quality, go to the `<SOURCE_BINARIES_DIRECTORY>` directory, and enter the Axon Agent installation parameters in the `SilentInput_AxonAgent.properties` file.
3. Log in to the machine as the Axon Administrator.
4. Ensure that no Axon-related service is running, and uninstall any previous version of Axon.
5. From the command line, run the `silentinstall.sh` file.

   The installer installs Axon and displays a message when the installation is complete. If you choose to install the Axon Agent, the installer displays a message when the Axon Agent installation is complete.
6. Optionally, to view the status of the Axon installation, open the `<INSTALLATION_DIRECTORY>/Configure.log` file. If the installation fails, open the log file `<user_home_directory>/silentErrorLog.log` to view the details of the error.
7. Optionally, to verify that the Axon Agent is running, open the `<Axon_Agent_Home>/logs/dq-axon/dqagent.log` file.

CHAPTER 5

After You Run the Installer

This chapter includes the following topics:

- Configure SSL Certificates for the Message Broker Service, 48
- Disable or Re-enable Segmentation, 49
- Configure Access to Enterprise Data Catalog, 49
- Connect to an SSL-Enabled Enterprise Data Catalog Instance, 51
- Configure Access to Informatica Data Quality, 52
- Configure Access to Data Privacy Management, 52
- Configure Axon with Operating System Profiles Enabled User for Data Integration Service, 53
- Connect to External Change Request Systems, 54
- Update Cache with Configuration Values, 59
- Configure SSL-certified User Authority for Axon, 59
- Check the Status of the Services, 60
- Log In To Axon, 61
- Apache Log4j RCE Vulnerabilities for Axon, 62

Configure SSL Certificates for the Message Broker Service

For the message broker service utilized by Axon to be fully functional, ensure that the SSL certificates are still valid.

You can find the SSL certificates in the following locations:

- `<Axon_install_dir>/certs/cacert`
- `<Axon_install_dir>/certs/server`

To verify the validity of the certificates, run the following command in the directory locations provided above:

```
openssl x509 -noout -text -in <cert_name> |grep -i -A2 validity
```

You can also confirm the certificate validity by reviewing the `Configure.log` file present in `<INSTALLATION_DIRECTORY>`. If the certificates have expired, the `Configure.log` file displays the following error: If the file displays the following error when you open it, then it means the certificates have expired:

```
MQPChannelHelper is not initialized. Updates to Lineage consumer will not be sent.
```

If the certificates for the message broker service have expired, replace the expired certificates with the latest certificates, by installing the fix **EBF-22417**.

To install **EBF-22417**, perform the following steps:

1. Navigate to the Technical Support File Transfer Protocol (TSFTP) server location at the following directory:
   `/updates/Axon/7.2.0/`
2. Download and extract the **EBF-22417** installer file to the Axon installation directory.
3. To launch the installer, run the following command:
   `./installEBF.sh`

Disable or Re-enable Segmentation

After you install or upgrade Axon, segmentation is automatically enabled, and you can restrict Axon objects by various segments. To make all Axon objects available to all Axon users, disable segmentation.

You must have the Super Admin profile to perform this task.

If you want to disable segmentation after the segments are created, delete all segments, and then disable segmentation.

1. To disable segmentation, go to the `<INSTALLATION_DIR>/bin` directory in the Linux environment, and run the following command:
   ```
   disable_authorization.sh
   ```
2. To enable segmentation again, go to the `<INSTALLATION_DIR>/bin` directory in the Linux environment, and run the following command:
   ```
   enable_authorization.sh
   ```

Configure Access to Enterprise Data Catalog

In Axon, you can configure Enterprise Data Catalog parameters from the Admin Panel. You must have the Super Admin profile to perform this task. After you configure the parameters, you can test the connectivity to Enterprise Data Catalog to ensure the entered parameters are correct.

After you install Axon, you can configure Enterprise Data Catalog parameters from the Admin Panel in the following scenarios:

- You did not select advanced configuration during Axon installation.
- You want to update the connection settings for Enterprise Data Catalog after the installation.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Enterprise Data Catalog**.

4. Click **Edit**, and configure the following properties:

Property	Description
Enterprise Data Catalog Server Host	URL of the Enterprise Data Catalog service in the following format: `http(s)://<host_name>` or `http(s)://<IP_address>`
Enterprise Data Catalog Server Port	Port number of the Enterprise Data Catalog service.
Enterprise Data Catalog Server Login User Name	User name to log in to Enterprise Data Catalog server.
Enterprise Data Catalog Server Login Password	Password to log in to Enterprise Data Catalog server.
Enterprise Data Catalog Server Login Namespace	User security domain of Enterprise Data Catalog server
Enterprise Data Catalog Resource Name	Name of the resource created in Enterprise Data Catalog that connects to your Axon application.
Axon Super Admin Email	Required. The email address of the SuperAdmin user responsible for onboarding discovered objects.
Automatically Accept Lineage Recommendations	Create attribute links automatically based on the lineage recommendations from Enterprise Data Catalog. If you do not select this option, you can manually create links later from the lineage recommendation notifications.
Enable Lineage Recommendation Notifications	Send notifications for the recommended attribute relationships based on the lineage in Enterprise Data Catalog.
Default Glossary	The default glossary name that you want to associate with the objects onboarded from Enterprise Data Catalog. Axon uses this default glossary if you choose to not onboard objects with associated glossaries to a System object.
Automatically Delete Attribute Relationships	Automatically deletes attribute relationships that are no longer valid if a linked physical field is deleted from Enterprise Data Catalog. This deletion applies to attribute relationships that are automatically created based on the lineage recommendations from Enterprise Data Catalog.
View Custom Attributes for Physical Fields	View custom attributes for physical fields from Enterprise Data Catalog.
Restrict Object Access Based on Enterprise Data Catalog	Restrict access to resources in Axon based on the access permissions to the assets in Enterprise Data Catalog. If you do not select this option, you can access all the assets in Axon irrespective of the access permissions to the assets in Enterprise Data Catalog. The default value is "False." **Note:** The access permission for a user from Enterprise Data Catalog is retrieved based on the email address. Ensure that the email filtering option is enabled in Enterprise Data Catalog. For more information, refer to *Informatica 10.4.1 Enterprise Data Catalog User Guide* .
Configure Timeout Period for Enterprise Data Catalog API	The maximum time period in seconds that Axon waits for the Enterprise Data Catalog API to respond.

Property	Description
Enterprise Data Catalog Server Proxy Host	The proxy host name for the Enterprise Data Catalog server. For more information on configuring the proxy parameters, contact Informatica Global Customer Support.
Enable Onboarder Failure Email Notification	Activate an email alert for the SuperAdmin when a data onboarding job fails. Select the checkbox to enable notification to SuperAdmin for onboarding process failures.
Enterprise Data Catalog Server Proxy Port Number	The proxy port number for the Enterprise Data Catalog server. For more information on configuring the proxy parameters, contact Informatica Global Customer Support.
Resume Failed Onboarding Job	Resume the previously failed onboarding job from the point of failure instead of restarting it. Select this check box to resume the previously failed onboarding job.

5. To test the connectivity to Enterprise Data Catalog, click **Test Connection**.

   If all the parameters are correct, you can view that the connection is successful. If any parameter is incorrect, the connection fails and an error message appears.

   Note: You cannot test the connectivity for the Enterprise Data Catalog proxy parameters.
6. Click **Save**.
7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Connect to an SSL-Enabled Enterprise Data Catalog Instance

To connect Axon to an SSL-enabled instance of Enterprise Data Catalog, you must upgrade cURL and the related libraries after you install Axon. If you do not upgrade, objects might not appear in Axon facets..

You must have the following library versions running on your Linux system:

- cURL version 7.29.0 or higher
- Network Security Services (NSS) version 3.21 or higher

To check the existing versions of cURL and NSS libraries, run the following command:

```
curl --version
```

To upgrade the cURL and NSS libraries to the latest versions, run the following commands:

```
yum upgrade curl
yum upgrade nss
```

Configure Access to Informatica Data Quality

If you have installed the Axon Agent, you must specify the details of the Axon Agent installation in Axon. Axon uses the parameters you specify to connect to Informatica Data Quality. You can open profiles or scorecards in Informatica Analyst tool from Axon local data quality rules.

You must have the SuperAdmin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Data Quality**.
4. Click **Edit**, and configure the following properties:

Property	Description
Axon Agent Server Host	Host name of the system where you installed the Axon Agent.
Axon Agent Server Port	Port number on which the Axon Agent runs.
IDQ Instance URL	URL of the Analyst Tool. To create a clickable link that associates data quality rule with profiles and scorecards, enter the Analyst Tool URL followed by the slash (/) character. For example, enter the URL in the following format: `http(s)://<hostname>:<port>/analyst/`

5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Configure Access to Data Privacy Management

If you selected the advanced configuration option during Axon installation, you would have entered the configuration settings for Data Privacy Management. If you did not select advanced configuration during

Axon installation or if you want to update the connection settings for Data Privacy Management after the installation, go to the Axon Admin Panel.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Data Privacy Management**.
4. Click **Edit**, and configure the following properties:

Property	Description
Data Privacy Management Server Host	URL of the Data Privacy Management service in the following format: `http(s)://<host_name>` or `http(s)://<IP_address>`
Data Privacy Management Server Port	Port number of the Data Privacy Management service
Data Privacy Management Server Login User Name	User name to log in to the Data Privacy Management server
Data Privacy Management Server Login Password	Password to log in to the Data Privacy Management server
Data Privacy Management Server Login Namespace	User security domain of Data Privacy Management

5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Configure Axon with Operating System Profiles Enabled User for Data Integration Service

After you install or upgrade Axon, you can configure the OS profile enabled user with Data Integration Service. After you configure the profile, you can use the operating system profile in the Axon Agent to run Data Quality rules.

To update the operating system profile user name, choose any of the following options:

Start DQAgent with your operating system profile user credentials

1. In the command prompt, navigate to the following Axon Agent home location: `DQAGENT_HOME` directory

2. To start the Axon Agent with the operating system profile user name, run the following command:
   `sh startup.sh --dis.ospUsername=<OSP Username>`

Configure DQAgent with the updated `application-mrs-dis properties` file

1. Navigate to the following Axon Agent home location:
   `DQAGENT_HOME/conf/directory`
2. Edit the `application-mrs-dis.properties` file.
3. Update the `dis.ospUsername` parameter with the user name enabled with the Data Integration Service.
4. To start the Axon Agent, run the following command from the command prompt:
   `sh startup.sh`

Connect to External Change Request Systems

You can configure Axon to connect to external change request systems that you use in your organization. When you create or modify an object, Axon can create a ticket in the change request system and include the object stakeholders in the workflow. For example, if you create a new Policy object in Axon, the change request system can create a new ticket to track the inputs and approvals for the policy. When a Policy stakeholder participates in the workflow for Policy objects, the change request system tracks the object creation process and notifies the Policy stakeholder.

You must have the Admin or Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **DG Operating Model** category, click **Change Request Systems**.

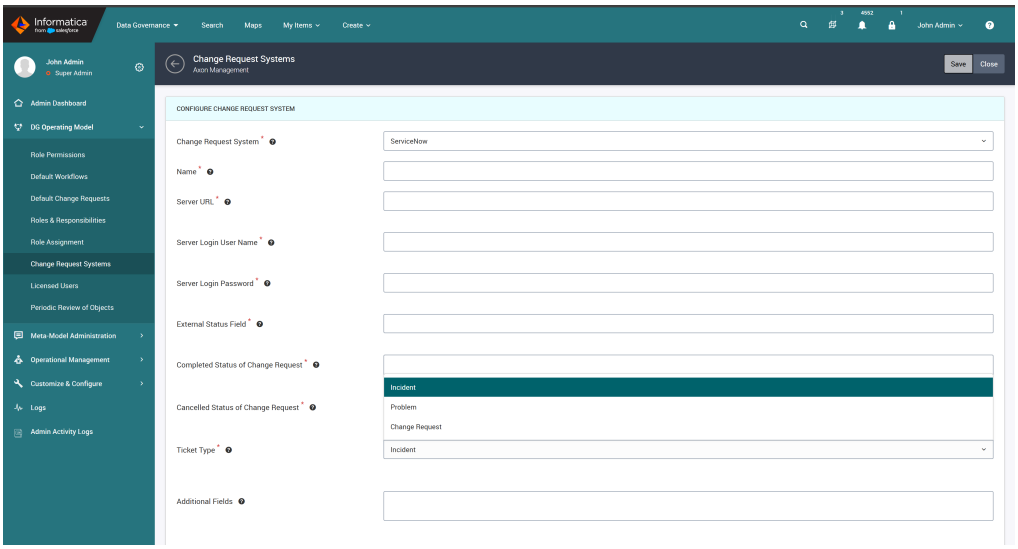
   The page displays the change request systems that you have configured. If you configure Axon to connect to several instances of an external change request system, the page displays all the instances.
3. Click on a change request system to configure its properties. Click **Create** to configure a new change request system for Axon to connect.

   The page displays the properties of the change request system.
4. Configure the following properties:

Property	Description
Change Request System	Select the change request system that you want to configure. Axon supports connection to ServiceNow and to Jira Cloud.
Name	Name of the change request system instance.
Server URL	URL of the change request system service in the following format: `http(s)://<host_name>` or `http(s)://<IP_address>`
Server Login User Name	User name to log in to the change request system
Server Login Password (ServiceNow only)	Password to log in to the change request system

Property	Description
Server Login Token (Jira Cloud only)	The token to log in to Jira Cloud. To get a server login token, log in to Jira Cloud, navigate to Account Settings > Security > Create
External Status Field	Enter the field name in the change requests system that indicates the ticket status. For example, if the change request system uses the "State" field to indicate the ticket status, enter `State` here.
Completed Status of Change Request	Enter the status in the **External Status Field** that corresponds to a completed change request. For example, if a completed status in the `State` field is indicated as "Resolved", enter `Resolved`.
Cancelled Status of Change Request	Enter the status in the change request system that corresponds to a cancelled change request. For example, enter `Cancelled` or `Revoked`.
Ticket Type	Select any one of the following ticket type based on your requirement: Incident: Track the status of an issue. Problem: Track the potential cause of one or more incidents. Change Request: Track the change request.
Project Key (Jira Cloud only)	Enter the key of the Jira Cloud project.
Issue Type (Jira Cloud only)	Enter the type of Jira Cloud issue. To get the Project Key, log in to Jira Cloud, navigate to Projects > View all Projects.
Additional Fields	Optional. Specify the additional fields that the change request system requires to create a ticket. Enter the values in JSON format. When you specify the fields, the fields appear in the **External Fields** section of an Axon change request. For example, if the ticketing system requires a change request to have a severity level and due date, you can configure `Severity` and `Due Date` as additional fields. For details on configuring the additional fields, see "Configure Additional Fields for External Change Request System" on page 56.

The following image shows an example of the Change Request System window:



5. Click **Save**.

Note:

- When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.
- If you connect Axon to Jira Cloud, Informatica recommends that do not create more than 100 issues at any given moment. If you do try to create more than 100 issues, the amount of time involved increases proportionally. Additionally, you cannot create more than 1000 issues.

Configure Additional Fields for External Change Request System

When you configure Axon to connect to an external change request system, the external system might require additional fields that do not appear by default in the Axon configuration page. Enter the details of the additional fields in JSON format so that Axon can create change requests in the external system.

Go to the Axon Admin Panel, and navigate to **DG Operating Model** > **Change Request Systems**. Select an external change request system, and configure the additional fields in the **Additional Fields** field.

Format

Add each additional field in the following format:

```
{
"key": "<field_name_in_external_system>",
"displayName": "<field_display_name_in_Axon",
"dataType": "<list_or_text_box>",
"defaultValue": "<default_value>",
"isMandatory": <required_or_not>,
"isHidden": <display_in_Axon_or_not>,
"isSensitive": <mask_field_or_not>
}
```

Enter square brackets `[` and `]` at the beginning and end of your entry. Separate each parameter by a a comma (,), and separate each field configuration by a comma (,).

Parameter Descriptions

The following table explains the parameters that you can enter for each field:

Parameter	Description
`key`	Name of the field in the external change request system. For example, if the field in the external system is "Urgency", enter the following parameter and value: `"key": "Urgency"`
`displayName`	Name of the field as you want it to appear on the Axon interface. For example, if you want the "Urgency" field in the external system to appear as **Importance** in Axon, enter the following parameter and value: `"displayName": "Importance"`

`dataType`	Specify whether the field must appear in Axon as a dropdown list or a blank text box.
	- To display the field in Axon as a dropdown list, enter the `list` value:
	`"dataType": "list"`
	- To display the field in Axon as a blank text box, leave the parameter as empty, or enter the `null` value:
	`"dataType": null`

`defaultValue`	Default values that you want to display for the field.
	- If you specify the field as a dropdown list in the `dataType` parameter, enter the dropdown values in a single line. Separate the values by a comma. For example, to specify the High, Medium, and Low values in the dropdown list, enter the following parameter and value:
	`"defaultValue": "High,Medium,Low"`
	- If you specify the field as a text box in the `dataType` parameter, enter the default text in the field. For example, to specify the "Accepted" default value in the text box, enter the following parameter and value:
	`"defaultValue": "Accepted"`

`isMandatory`	Specify whether the field is required or optional. For required fields, Axon appends an asterisk (*) to the field name.
	- To mark the field as required, enter the `true` value:
	`"isMandatory": true`
	- To mark the field as optional, enter the `false` value:
	`"dataType": false`

`isHidden`	Specify whether or not the field must appear in the Axon interface.
	- If the field is essential to create a ticket, and you do not want to display the field in the Axon interface, enter the `true` value:
	`"isHidden": true`
	- If the field is essential to create a ticket, and you want to display the field in the Axon interface, enter the `false` value:
	`"isHidden": false`

`isSensitive`	Specify whether you want to use asterisks to mask the values that the user enters in the field. For example, if the user enters an employee code, you might want to mask the entry.
	Note: To specify a field as sensitive, you must specify the field as a dropdown list in the `dataType` parameter.
	- To mask the user inputs, enter the `true` value:
	`"isSensitive": true`
	- To not mask the user inputs, enter the `false` value:
	`"isSensitive": false`

Example

You want to display the following fields from the external change request system in Axon:

Parameter	Field 1	Field 2	Field 3
"key"	The field in the external system is "Severity"	The field in the external system is "Description".	The field in the external system is "Category".
"displayName"	You want to display the field as **Importance** in Axon.	You want to display the field as **Description** in Axon.	You want to display the field as **Select the Issue Type** in Axon.
"dataType"	The field is a dropdown list.	The field is a text box.	The field is a dropdown list.
"defaultValue"	The dropdown values of the list are High, Medium, and Low.	The default entry in the text box is "Enter an optional comment."	The dropdown values of the list are Hardware, Software, and Other.
"isMandatory"	The field is mandatory.	The field is not mandatory.	The field is not mandatory.
"isHidden"	The field is not hidden in the Axon interface.	The field is not hidden in the Axon interface.	The field is not hidden in the Axon interface.
"isSensitive"	The field is not sensitive and does not need to be masked.	The field is not sensitive and does not need to be masked.	The field is not sensitive and does not need to be masked.

To configure the fields, enter the following values in the **Additional Fields** field:

```
[
{
"key": "Severity",
"displayName": "Importance",
"dataType": "list",
"defaultValue": "High,Medium,Low",
"isMandatory": true,
"isHidden": false,
"isSensitive": false
},
{
"key": "Description",
"displayName": "Description",
"dataType": null,
"defaultValue": "Enter an optional comment.",
"isMandatory": false,
"isHidden": false,
"isSensitive": false
},
{
"key": "Category",
"displayName": "Select the Issue Type",
"dataType": "list",
"defaultValue": "Hardware,Software,Other",
"isMandatory": false,
"isHidden": false,
"isSensitive": false
}
]
```

Update Cache with Configuration Values

Some Axon configurations in the Admin Panel are saved to the Axon database, but these configuration values are not loaded to the Axon cache. To make the configuration changes available to the Axon users, you must update the Axon cache with the new configuration values.

You must update the Axon cache when you perform the following configurations:

- Display the logo of your organization in the Axon toolbar
- Change the colors of the Axon interface to reflect the theme and colors of your organization
- Configure email settings
- Configure regional settings
- Configure glossary types in the rollup view
- Configure saved searches for quick links
- Configure email delivery options
- Configure a SAML 2.0 application to work with Axon
- Configure Access Control
- Configure access to Enterprise Data Catalog
- Configure access to Data Quality
- Configure access to Data Privacy Management
- Configure access to the LDAP server
- Enable dashboard
- Specify prefix for the reference numbers of Axon objects
- Send diagnostic information to Informatica Global Customer Support

To clear the Axon cache and load the cache with the new configuration values, run the following command in the Linux environment:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Memcached and HTTPD services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Configure SSL-certified User Authority for Axon

You must provide Axon users the necessary SSL authentication by a Certified Authority so that their browser access to Axon is secure.

1. Open the `informatica_web.conf` file available in the following location:

   `<INSTALLATION_DIR>/axonhome/third-party-app/httpd/conf.d/`
2. Uncomment the `SSLCaCertificateFile` property and enter the full path to your SSL CA file.

Check the Status of the Services

You can verify if all the services installed by Axon are running properly.

Log in to Axon as an administrator, go to the **Admin Panel**, and view the **Admin Dashboard**. The **Status of Axon Services** section shows the services that are operational.

Axon uses the following services, which are located in the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment:

Service	Service Name	Service Script
Authorization	`authorization`	`axon_authorization_microservice_init`
Bulk Upload	`axon_bulk_upload_microservice`	`axon_bulk_upload_microservice_init`
Change Request	`axon_changerequest_microservice`	`axon_changerequest_microservice`
Daemon Tools	`axon_daemontools`	`daemontools`
Data Marketplace	`axon_marketplace_microservice`	`axon_marketplace_microservice_init`
Document Upload	`axon_document_upload_microservice`	`axon_document_upload_microservice`
Email Notification	`jobber`	`jobber`
Event Monitor	`enable_eventmonitor`	`enable_eventmonitor.sh`
Graph Database and Search	`orientdb`	`orientdb`
HTTPD	`httpd`	`httpd`
Log Search	`elasticsearch`	`elasticsearch`
Manage Logs	`Logstash`	`logstash`
Memcached	`memcached`	`memcached`
Messaging	`rabbitmq-server`	`rabbitmq`
Notification	`axon_notification_microservice`	`axon_notification_microservice`
OdbConsumer	`OdbConsumer`	`OdbConsumer`
PostgreSQL	`postgresql`	`postgresql`
Unison Search	`axon_unison_microservice`	`axon_unison_microservice_init`
Value List	`axon_value_list_microservice`	`axon_valuelist_microservice_init`
Workflow	`camunda`	`camunda`

To check the status of a specific service, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory, and run the following command:

```
sh <service_script> status
```

For example, to check the status of the Email Notification service, run the following command:

```
sh jobber status
```

To check the status of all Axon services, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory, and run the following command:

```
sh axonStatus
```

Note: If the PostgreSQL database is deployed on external instances, such as Amazon RDS and Microsoft Azure for PosgreSQL Databases, you need to contact the database administrator for the status of services.

Log In To Axon

To log in to Axon from a browser, use the URL displayed by the Axon installer after the installation completes. To configure Single Sign-On for authentication, see the *Configure Single Sign-On* topic in the Axon Data Governance Administrator Guide.

The Axon installer assigns the following default port numbers for the Apache server:

- If you use HTTP authentication, the installer assigns 9999 as the default port number for the Apache server.
- If you use HTTPS authentication, the installer assigns 9443 as the default port number for the Apache server.

You can also enter a custom port number for the Apache server. Depending on whether you selected the default port number or entered a custom port number for the Apache server during installation, you must enter the correct port number in the browser address field.

1. Open a browser.
2. In the address field, type the fully qualified host name for the Axon login page in the following format: `http(s)://<host_name>:<9999_or_9443_or_custom_port_number>/`
3. From the Axon home page, on the top right corner of the toolbar, click **Login**.
4. Enter your login credentials. The following are the default login credentials for the Super Admin user:
   Username: `admin@informatica.com`

   Password: `Changeme@123`
5. Click **Login**.
   Axon refreshes the page and displays your username in the toolbar, indicating that you have successfully logged in.

When you log in to Axon for the first time, Informatica recommends that you modify your login password for security purposes. The password must be at least eight characters long and contain at least one lowercase character, one uppercase character, one numeric digit, and one special character.

Apache Log4j RCE Vulnerabilities for Axon

The Log4j vulnerability has been addressed for Axon. You might find earlier versions of Log4j in the Axon and Axon Agent. The Axon backup folders contain earlier versions of Log4j vulnerabilities. These backup folders are not used by Axon. You can safely ignore or remove these backup folders.

If you have the Axon Agent connected to a version of Informatica Data Quality earlier than 10.5.2, you will find the Log4j vulnerability in the earlier version of Axon library files. We recommend you install the Axon Agent for Data Quality version 10.5.2 in order to remove the vulnerability.

Note: Backup folders are required only to revert to an earlier version of Axon. If you do not need to roll back to a previous version of Axon, you can delete the backup folders. The latest Axon 7.3 version has Log4j fixes included.

CHAPTER 6

Uninstallation

This chapter includes the following topics:

- Running the Axon Uninstaller, 63
- Running the Axon Agent Uninstaller, 64

Running the Axon Uninstaller

Run the uninstaller file to remove all the Axon files present in the Axon home and installation directories. The uninstaller script stops the Axon services that are currently running, and removes all the Axon and third-party packages and tar files from the system.

1. Log in to the Linux system where you have installed Axon.
2. Go to the `<INSTALLATION_DIR>/Uninstaller_axon` directory and run the following command:

   ```
   uninstaller.sh
   ```
3. Review the uninstallation summary file at the following location: `<user_home_directory>/Uninstall.log`
4. Press the **Enter** key to complete the uninstallation.
5. Run the following command to delete the uninstaller directory:

   ```
   rm -rf <INSTALLATION_DIR>/Uninstaller_axon
   ```
6. Check whether all services have stopped to verify that uninstallation is complete. For more information, see the *Working With Services* topic in the *Monitoring Axon* chapter of the *Axon Data Governance Administrator Guide*.

   Note: If the `<INSTALLATION_DIR>/Uninstaller_axon` directory does not contain the `uninstaller.sh` script file, perform the following tasks:

   1. Go to the `<INSTALLATION_DIR>/bin` directory and run the following command to shut down Axon:

      ```
      shutdown.sh
      ```
   2. Delete the `<INSTALLATION_DIR>` directory where Axon is installed.

Note:

1. If you have deployed the PostgreSQL database on an external node, the uninstaller does not remove the contents of the database in the deployment directory. Go to the deployment directory at the external node to manually remove the contents of the PostgreSQL database.

2. If you have deployed the graph database in a cluster on external nodes, the uninstaller does not remove the cluster on the nodes. You need to run the cleanup script separately to uninstall the graph database cluster. For more information, see the *Axon uninstaller does not uninstall the graph database cluster* article on the Knowledge Base site.

Running the Axon Agent Uninstaller

Run the Axon Agent uninstaller to remove the Axon Agent from your system.

1. Navigate to the following directory

   `<AXONAGENT_INSTALLATION>/Uninstaller_agent`
2. Run the `uninstaller.sh` file.
3. In the **Uninstallation** step, review the information and press the **Enter** key to continue with the uninstallation of the Axon Agent.
4. In the **Uninstalling** step, review the progress of the uninstallation.
5. In the Post-Uninstallation Summary screen, review the information.
6. Press the **Enter** key to complete the uninstallation.

Note: If you installed the Axon Agent in silent mode, the uninstaller removes the Axon Agent in silent mode. The console does not prompt you to press the **Enter** key after you run the uninstaller file, and it does not display the progress indicator during uninstallation.

CHAPTER 7

Troubleshooting

This chapter includes the following topics:

- Troubleshooting Overview, 65
- Services Fail to Start After Installation, 65
- Cannot Log In to Axon, 66
- Cannot Start the HTTPD Service because SSL is Password-Protected, 67
- Axon Does Not Load in the Browser, 68
- Incorrect Host Name, 69
- Validation Utility Errors, 69
- Uninstallation is Incomplete, 70
- Warnings in Configure.log Files, 70
- Installation errors, 70

Troubleshooting Overview

If you encounter errors when you install Axon Data Governance, try the following solutions before you contact Informatica Global Customer Support.

Services Fail to Start After Installation

After the Axon installation is complete, the Axon application and all services are launched. Sometimes, some services fail to start when Axon is started.

To see the status of services when Axon starts, see the "Check the Status of the Services" on page 60 topic. If the Messaging, OdbConsumer, or Graph Database service fails to start when Axon is started, perform the following tasks to troubleshoot the issue.

Verify the IP Address and Domain Name of the Axon Instance

To verify that the IP address and domain name of the Axon instance is correctly configured in the Linux environment, perform the following steps:

1. Run the following command to see the IP address of the Axon instance:

```
ifconfig
```

2. Run the following command to see the fully qualified domain name of the Axon instance:

   ```
   hostname -f
   ```

3. Run the following command to see the network address of the Axon host:

   ```
   hostname -i
   ```

   The network address of the Axon host must match the IP address of the Axon host. If the addresses match, move to the next step.

4. Open the `/etc/hosts` file and make sure that the correct IP address and fully qualified domain name is mentioned in the file.

5. Go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

   ```
   sh startup.sh
   ```

Remove the Service PID File

If the issue persists after the above troubleshooting steps, you need to delete the process identification number (PID) file for the service. Perform the following steps in the Linux environment:

1. Go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to shut down the Axon instance:

   ```
   sh shutdown.sh
   ```

2. Delete the process ID file for the service that fails to start.
   - For the Messaging service, delete the `<INSTALLATION_DIR>/axonhome/third-party-app/run/rabbitmq/rabbitmq.pid` file.
   - For the OdbConsumer service, delete the `<INSTALLATION_DIR>/axonhome/third-party-app/run/OdbConsumer.pid` file.
   - For the Graph Database service, delete the `<INSTALLATION_DIR>/axonhome/third-party-app/run/orientdb.pid` file.

3. Check for other service PID files that might have been created by users that are not Axon Administrators. Delete the PID files.

4. Go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

   ```
   sh startup.sh
   ```

Cannot Log In to Axon

If your Axon license expires, users cannot log in to Axon. Logged in users automatically log out of Axon at the time of license expiry.

Perform the following tasks if your Axon license expires:

Verify that the PostgreSQL Database Service is Running

To verify that the PostgreSQL database service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh postgresql status
```

If the service is shut down, go to the <INSTALLATION_DIR>/bin directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Note: Use the same commands to verify that the database services is running for the PostgreSQL database deployed on a different node and external nodes, such Amazon RDS and Microsoft Azure Database for PostgreSQL.

Replace Your Axon License Key

To replace your Axon license key, perform the following steps:

1. Contact Informatica Global Customer Support to obtain a new license key.
2. In the Linux environment, replace the expired license key file with the new license key file in the following directory:

   <INSTALLATION_DIR>/axonhome/axon/

   Note: Make sure that the new license key file and expired license key file share the same name.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Cannot Start the HTTPD Service because SSL is Password-Protected

If your SSL certificate is protected by a password, the HTTPD service cannot start. You are prompted for the password each time the HTTPD service is started or restarted.

To avoid entering the password each time the HTTPD service starts, perform the following steps:

1. Create a new .conf file to store your SSL password.
   For example, you can create a password.conf file in the <INSTALLATION_DIR>/axonhome/third-party-app/httpd/ directory.
2. In the SSL password file you created, add the following lines to store your SSL password:

   ```
   #!/bin/sh
   echo "<your_SSL_password>"
   ```

   For example, add the following lines to the <INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf file:

   ```
   #!/bin/sh
   echo "Qwpt123%"
   ```
3. Run the following command to provide the execute permission to the SSL password file:

   ```
   chmod +x <path_to_file>
   ```

   For the sample file <INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf created in Step 1, run the following command:

   ```
   chmod +x <INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf
   ```

4. Modify the SSL certificate file at `<INSTALLATION_DIR>/axonhome/third-party-app/httpd/conf.d/ssl.conf` to point applications to refer to the SSL password file you created. Change the `SSLPassPhraseDialog builtin` parameter to `SSLPassPhraseDialog exec:<path_to_your_SSL_password_file>`.
   For the sample file `<INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf` created in Step 1, open the `ssl.conf` file and change the `SSLPassPhraseDialog builtin` parameter to `SSLPassPhraseDialog exec:/<INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf`.

   After you make the changes, the Axon HTTPD service retrieves the SSL password from the `password.conf` file instead of the `ssl.conf` file.

Axon Does Not Load in the Browser

You can access Axon by typing the fully qualified host name in the address bar of a web browser. In the address field, type the fully qualified host name for the Axon login page in the following format: `http(s)://<host_name>:<port_number>/`

If Axon does not load in the browser, perform the following tasks:

Verify the Firewall Settings

You must enable the firewall of the machine where you installed Axon to allow requests for the Axon port.

Verify that the HTTPD Service is Running

To verify that the HTTPD service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh httpd status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the Unison Search Service is Running

To verify that the Unison Search service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh axon_unison_microservice_init status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify Log Files

Verify the following log files:

- `<INSTALLATION_DIR>/axonhome/third-party-app/httpd/logs/<hostname>-error_log` for HTTP authentication
- `<INSTALLATION_DIR>/axonhome/third-party-app/httpd/logs/<hostanme>-ssl_error_log` for HTTPS authentication
- `<INSTALLATION_DIR>/axonhome/axon/var/logs/prod.log`

Verify Your Axon License Validity

Check if your Axon license has expired. If your license expires, users cannot log in to Axon. Perform the following tasks if your Axon license expires:

1. Contact Informatica Global Customer Support to obtain a new license key.
2. In the Linux environment, replace the expired license key file with the new license key file in the following directory:

   ```
   <INSTALLATION_DIR>/axonhome/axon/
   ```

   Note: Make sure that the new license key file and expired license key file share the same name.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Verify Axon Settings and Parameters

Check if the Axon settings and **Admin Panel** parameters are correct. You can modify some of the **Admin Panel** settings from the Linux environment. You can run specific commands to display the current parameter values, reset the parameters to their default values, or enter new parameter values.

For more information, see the *Modify Admin Panel Settings from the Linux Environment* topic in the *Monitoring Axon* chapter of the Axon Data Governance Administrator Guide.

Incorrect Host Name

If you encounter an error with the host name during installation, verify that the host name has been configured with the correct IP address.

Run the following command to show the IP address of your host name:

```
hostname -i
```

Run the following command to show the fully qualified domain name of your host:

```
hostname -f
```

Validation Utility Errors

You might see errors relating to some package libraries when you run the validation utility.

Upgrade your version of OpenSSL to openssl-libs-1.0.2k if you see the following error when you run the validation utility:

```
Your system contains internal RPMs that are lower than the versions required for Axon.
Install the following internal RPM versions before you proceed: openssl-libs-1.0.2k
```

Uninstallation is Incomplete

If Axon is not uninstalled properly, the services are not stopped. The `Uninstaller_axon` directory is deleted from your system, but other Axon files and directories remain.

This issue occurs if you did not press the **Enter** key after Step 7 of the Axon installation procedure. For more information, see "Installing Axon" on page 32.

Perform the following steps to complete the uninstallation and remove all the Axon files present in the Axon home and installation directories:

1. Go to the `<INSTALLATION_DIR>/bin` directory and run the following command to shut down Axon:

   ```
   shutdown.sh
   ```

2. Delete the `<INSTALLATION_DIR>` directory where Axon is installed.

To verify that uninstallation is complete, check whether all services have stopped. If any service is running, you must stop the service. For more information, refer to "Check the Status of the Services" on page 60.

Warnings in Configure.log Files

After you install Axon, you might find a few warnings in the `Configure.log` file due to the previous RabbitMQ and Erlang versions.

You can stop running the previous versions of the RabbitMQ and Erlang libraries, or upgrade the shared libraries. If you plan to upgrade the shared libraries, use the following versions:

- rabbitmq_server-3.8.3
- erlang-22.3

Verify that the host name and IP addresses are configured correctly on the server.

Installation errors

You might see the following error in the Step 2 of 8 during Axon installation:

```
ERROR: Python is not found
```

Perform the following steps in your Linux environment to fix the error:

- Enter `which python` to find the python versions that are installed. For example, the server may have the following pre-installed python versions:

  - /usr/bin/python27
  - /usr/bin/python36

- To create a symlink as `python`:

  - If you have root access, run the following command:

    ```
    ln -s /usr/bin/<python_version> /usr/bin/python
    ```

    For example, you can enter: `ln -s /usr/bin/python2.7 /usr/bin/python`

- If you do not have root access, run the following command:
  `ln -s /usr/bin/<python_version> /home/<NON_ROOT_USER>/python/bin/python`

  For example, you can enter: `ln -s /usr/bin/python2.7 /home/<NON_ROOT_USER>/python/bin/python`

- Run the following command to set the environment variable in Linux:
  `export PATH=$PATH:/home/<NON_ROOT_USER>/python/bin`

INDEX

A

administrator 13

D

data collection technical owner 13
data consumer 13
Data Marketplace technical owner 13
data owner 13

S

segmentation
  disable 49
  re-enable 49
services
  check status 60
SuperAdmin 13

U

user profile
  admin 11
  superadmin 11
  webuser 11
user types
  administrator 13
  data collection technical owner 13
  data consumer 13
  Data Marketplace technical owner 13
  data owner 13
  SuperAdmin 13